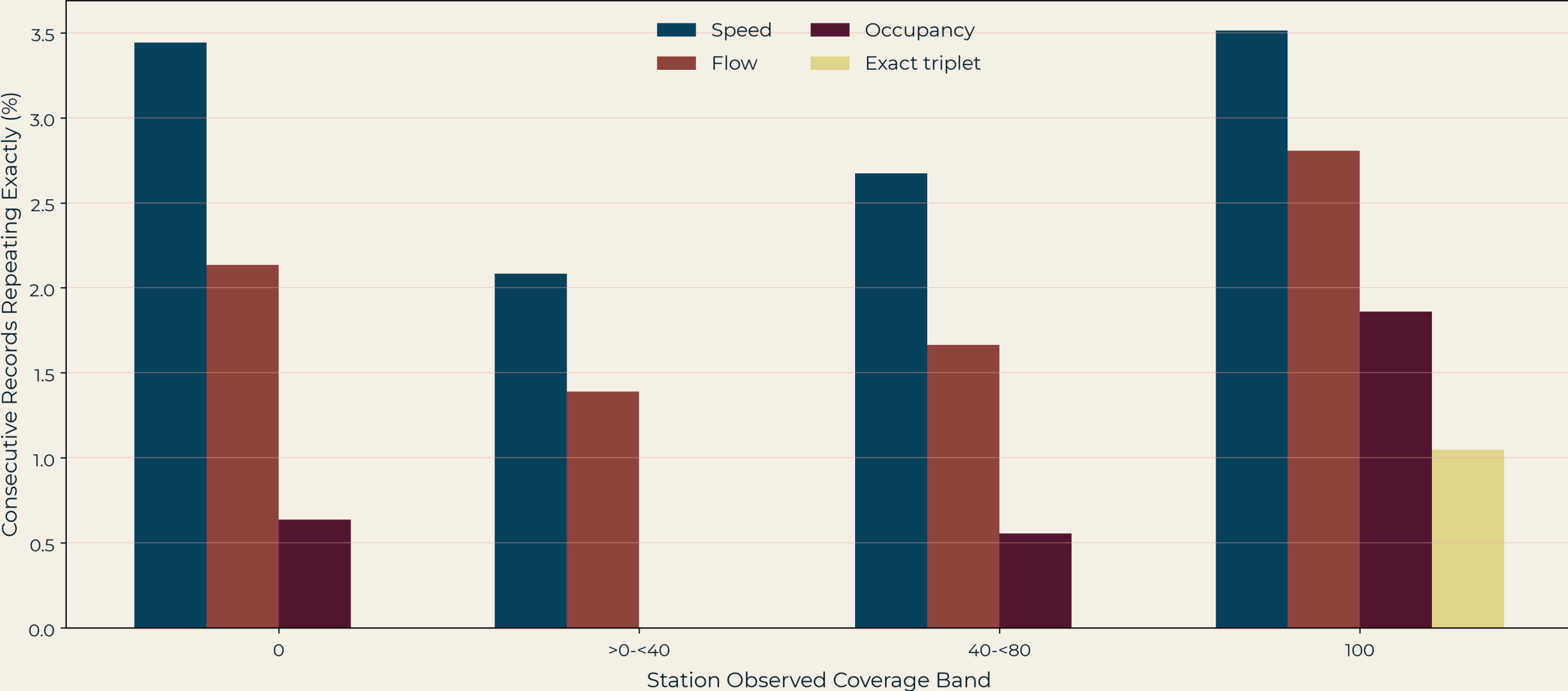


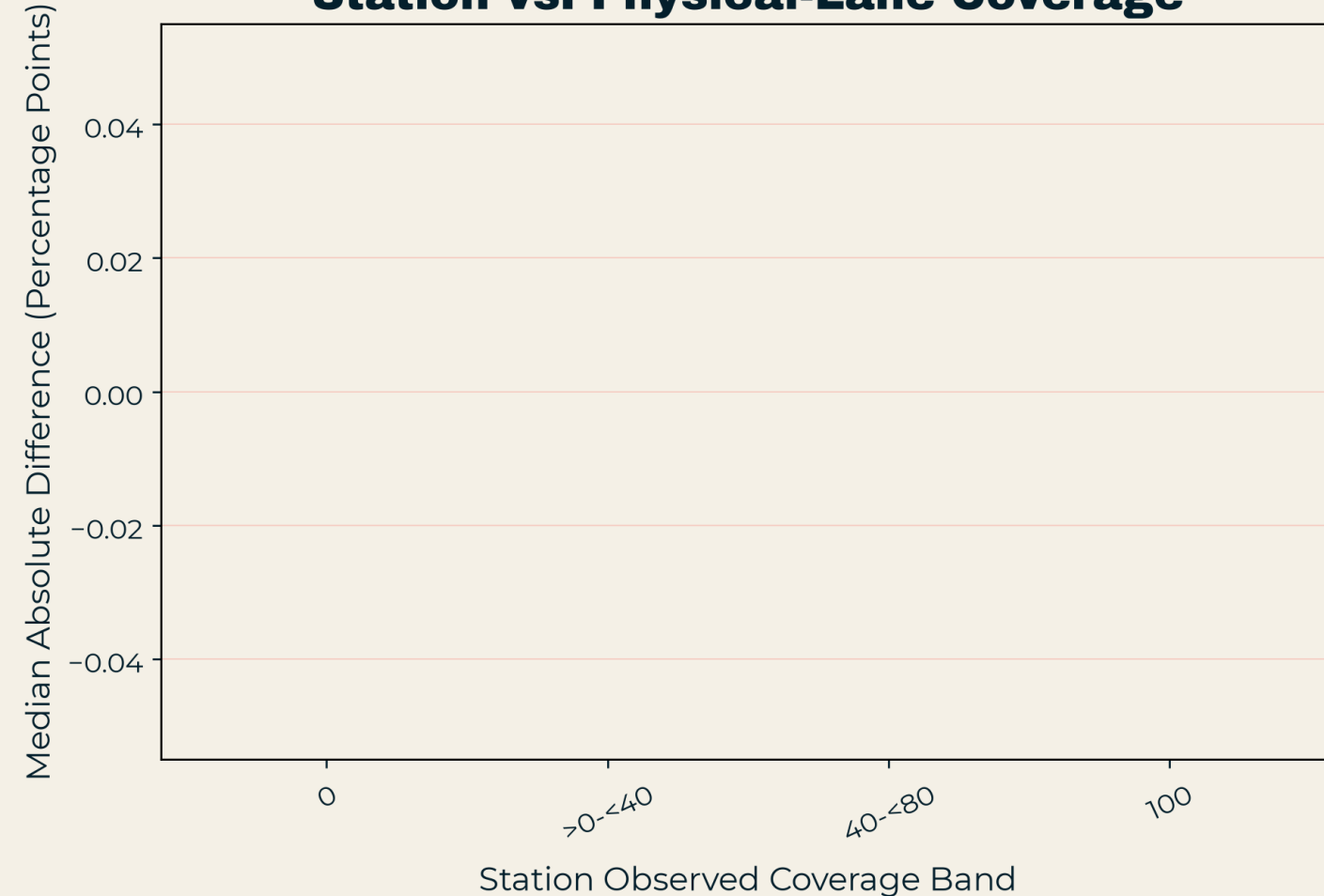
# Exact Repeats Do Not Spike in the 0%-Observed Rows



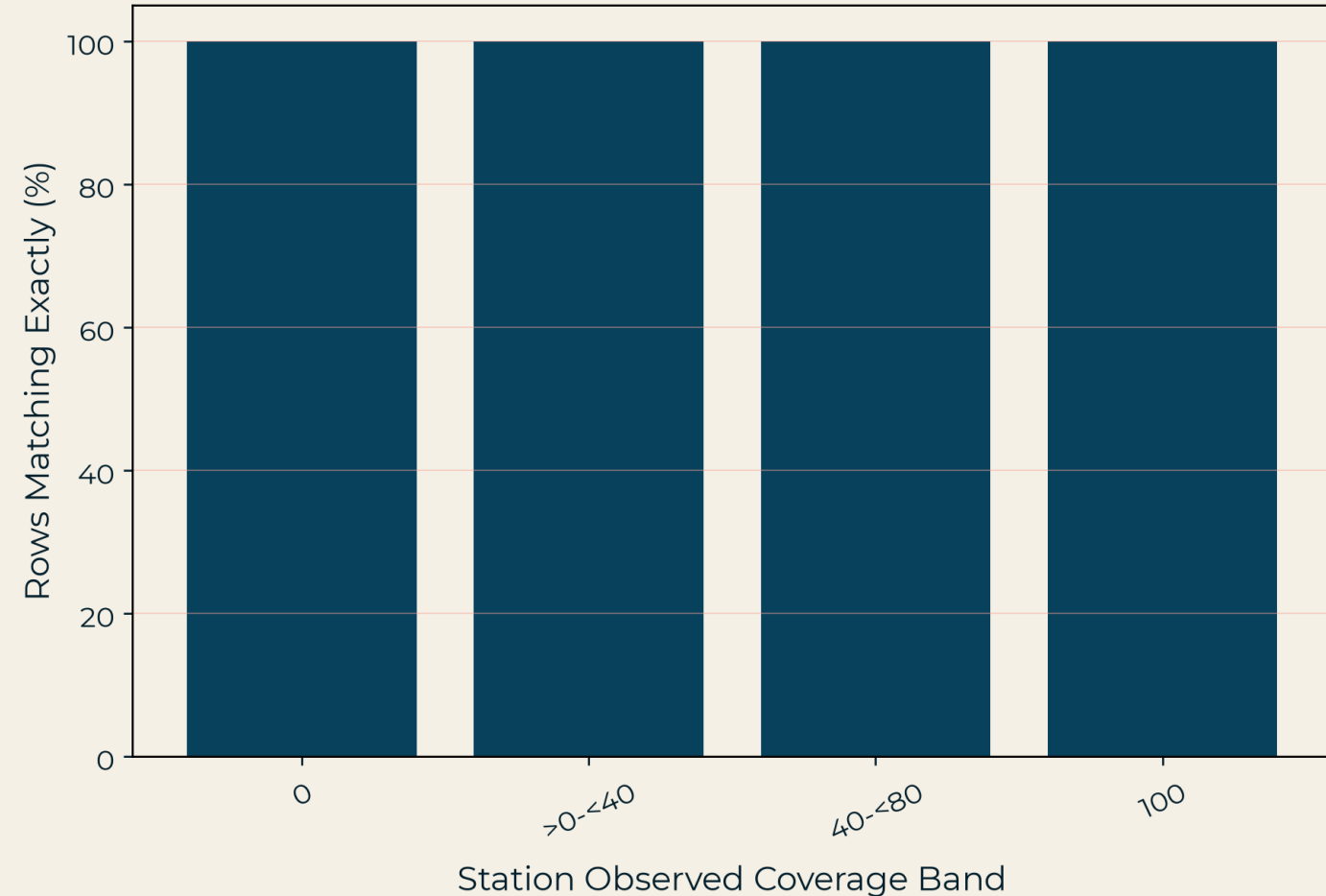
Exact repeats are a diagnostic for suspicious persistence, not proof of imputation.

# Lane Audit Uses Metadata-Defined Physical Lanes, Not Padded Slots

## Station vs. Physical-Lane Coverage

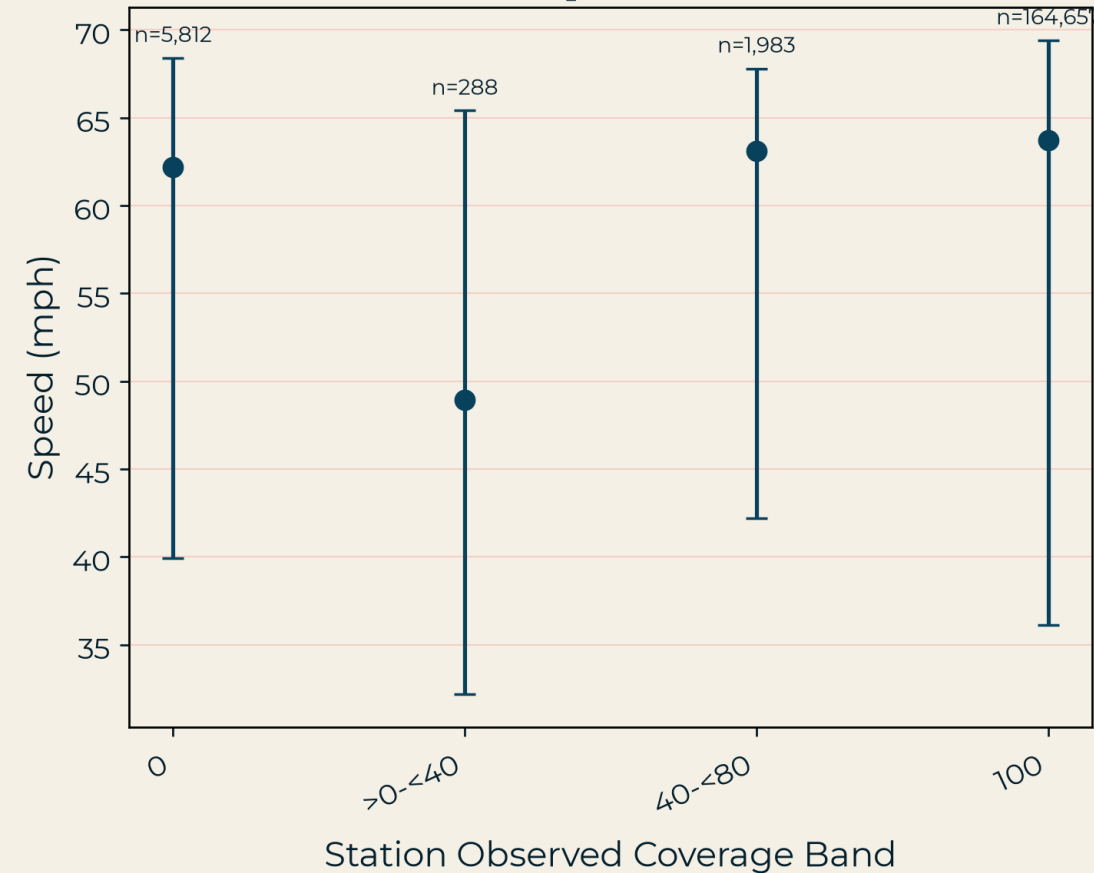


## Exact Station/Lane Agreement

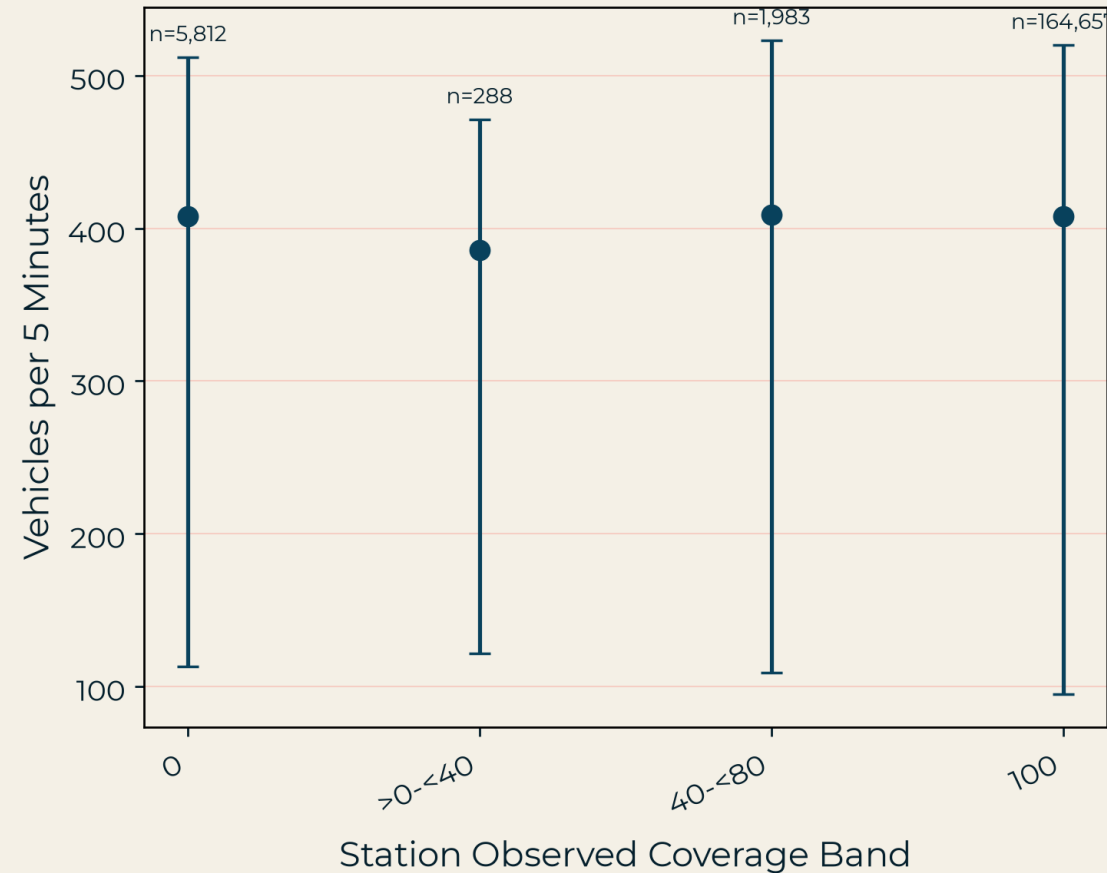


# Traffic Measurements Remain Populated Even When Coverage Is Zero

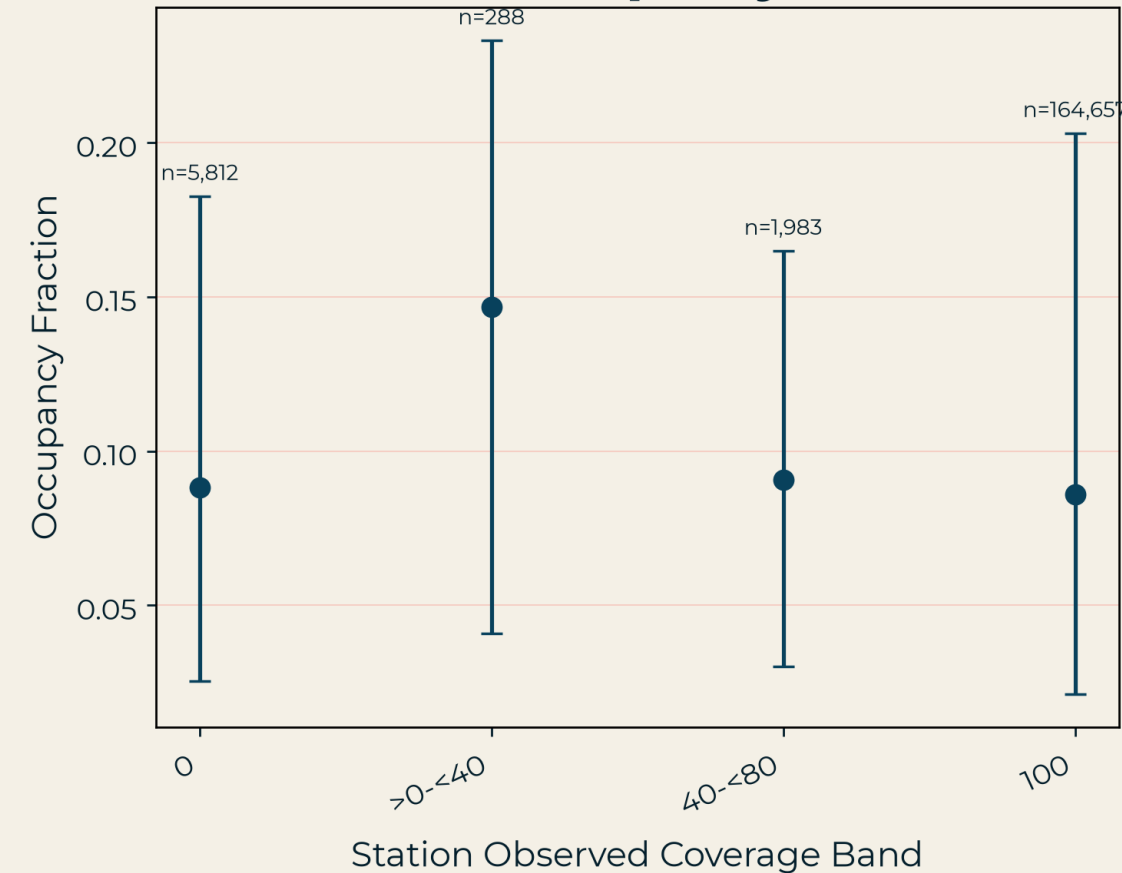
## Speed



## Flow



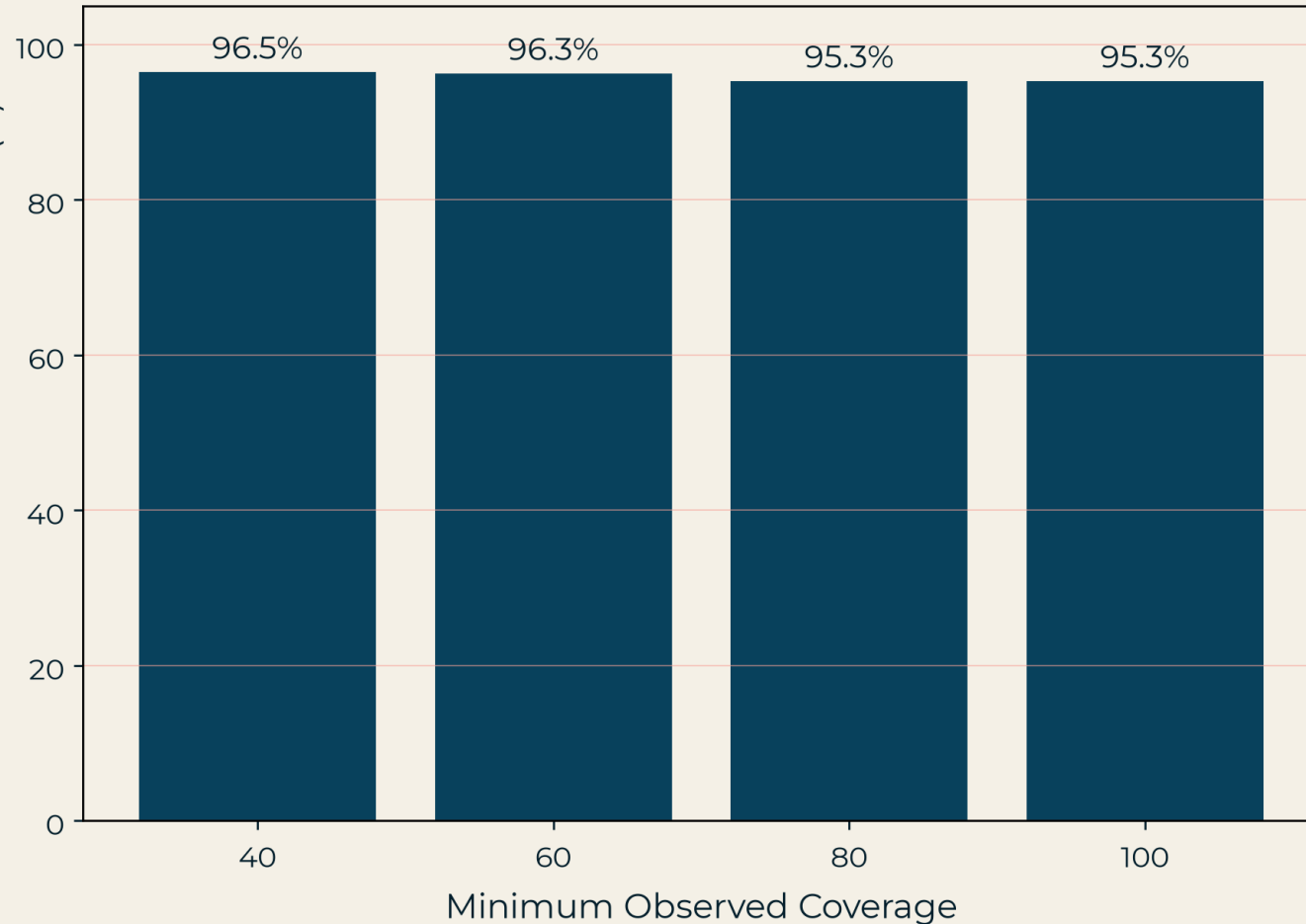
## Occupancy



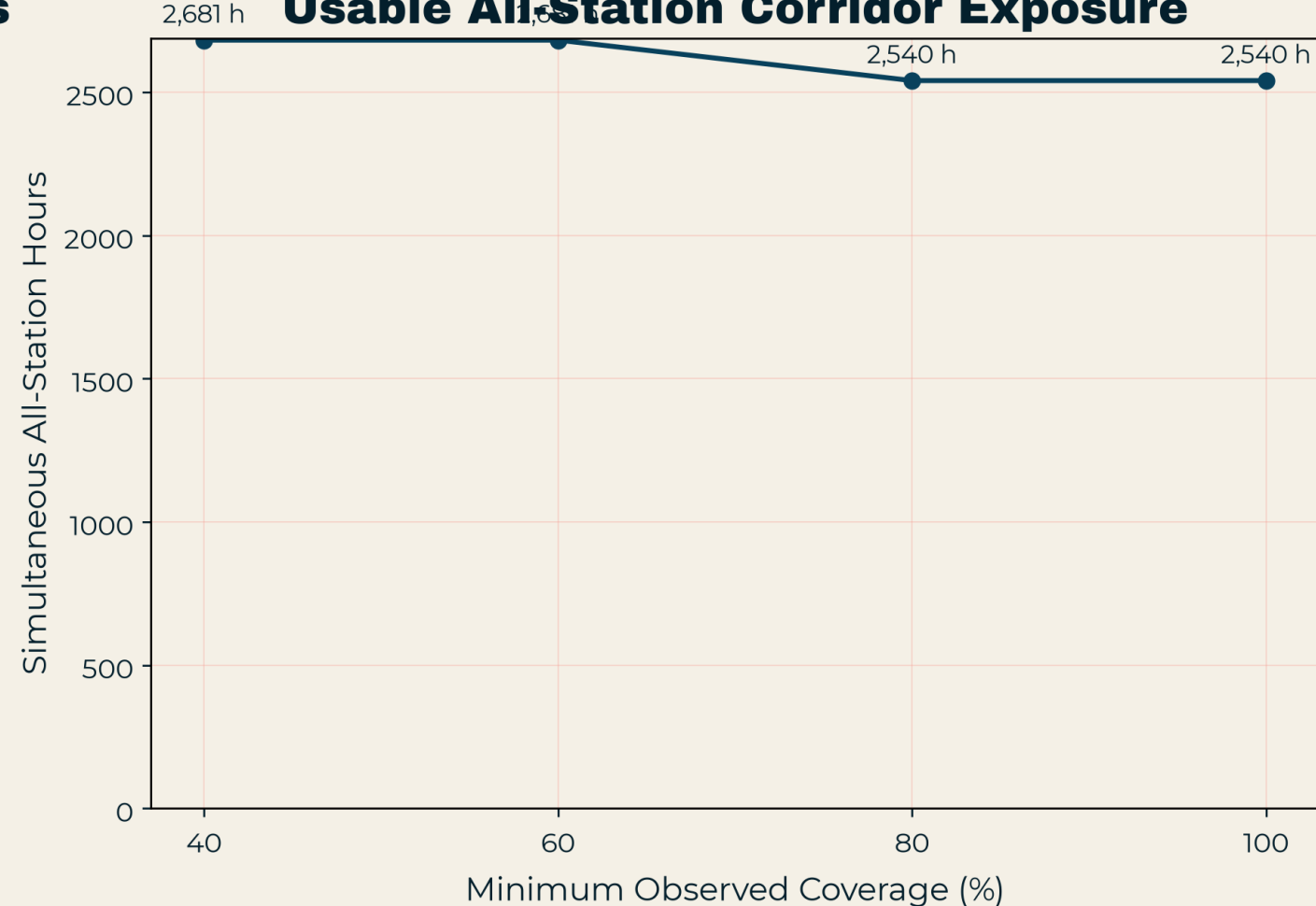
Points are medians; bars span the 10th–90th percentiles. Populated values do not make 0%-observed rows analytically valid.

# Coverage Threshold Sensitivity — I-10 East Primary Corridor

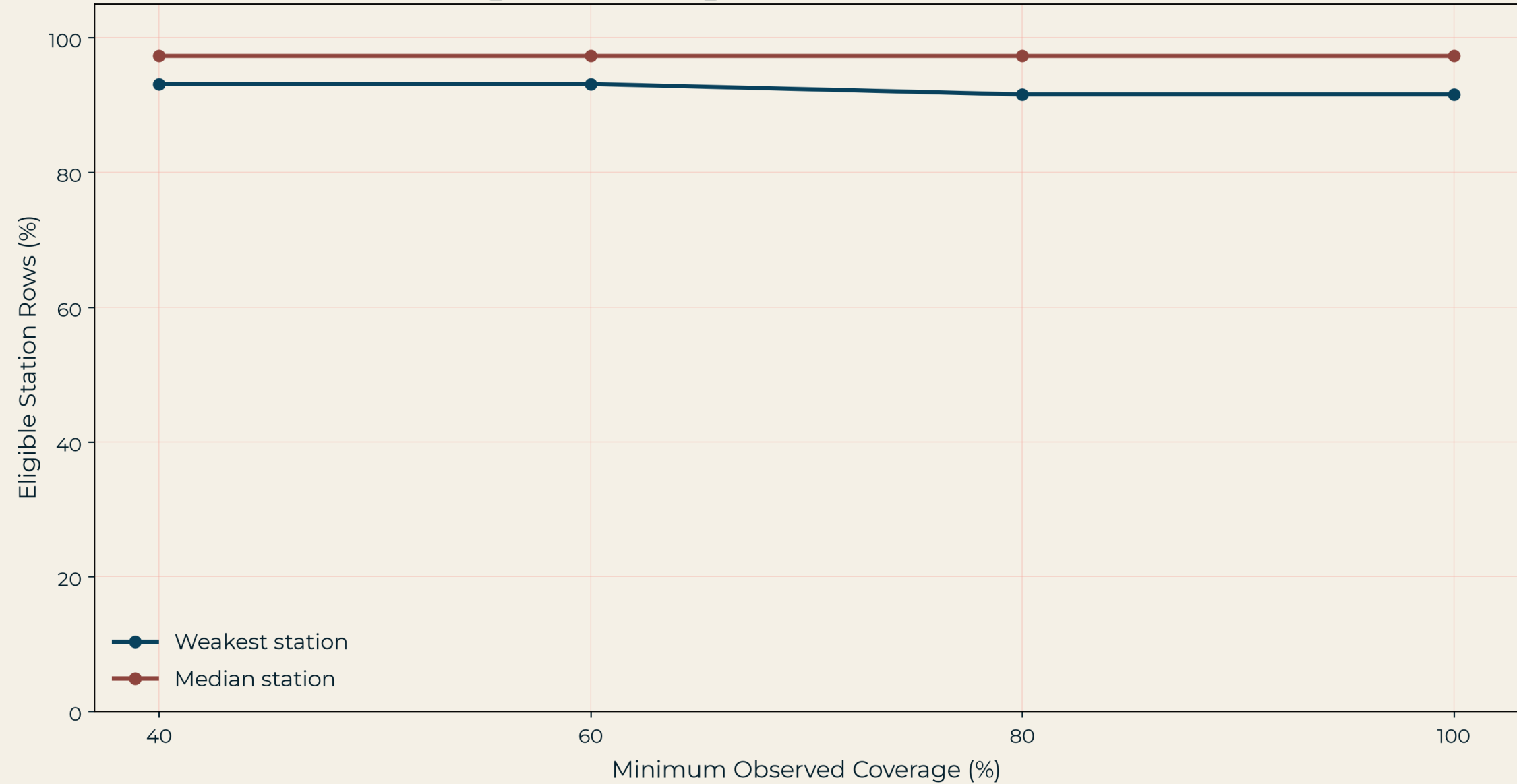
## Data Retained as Coverage Requirement Tightens



## Usable All-Station Corridor Exposure

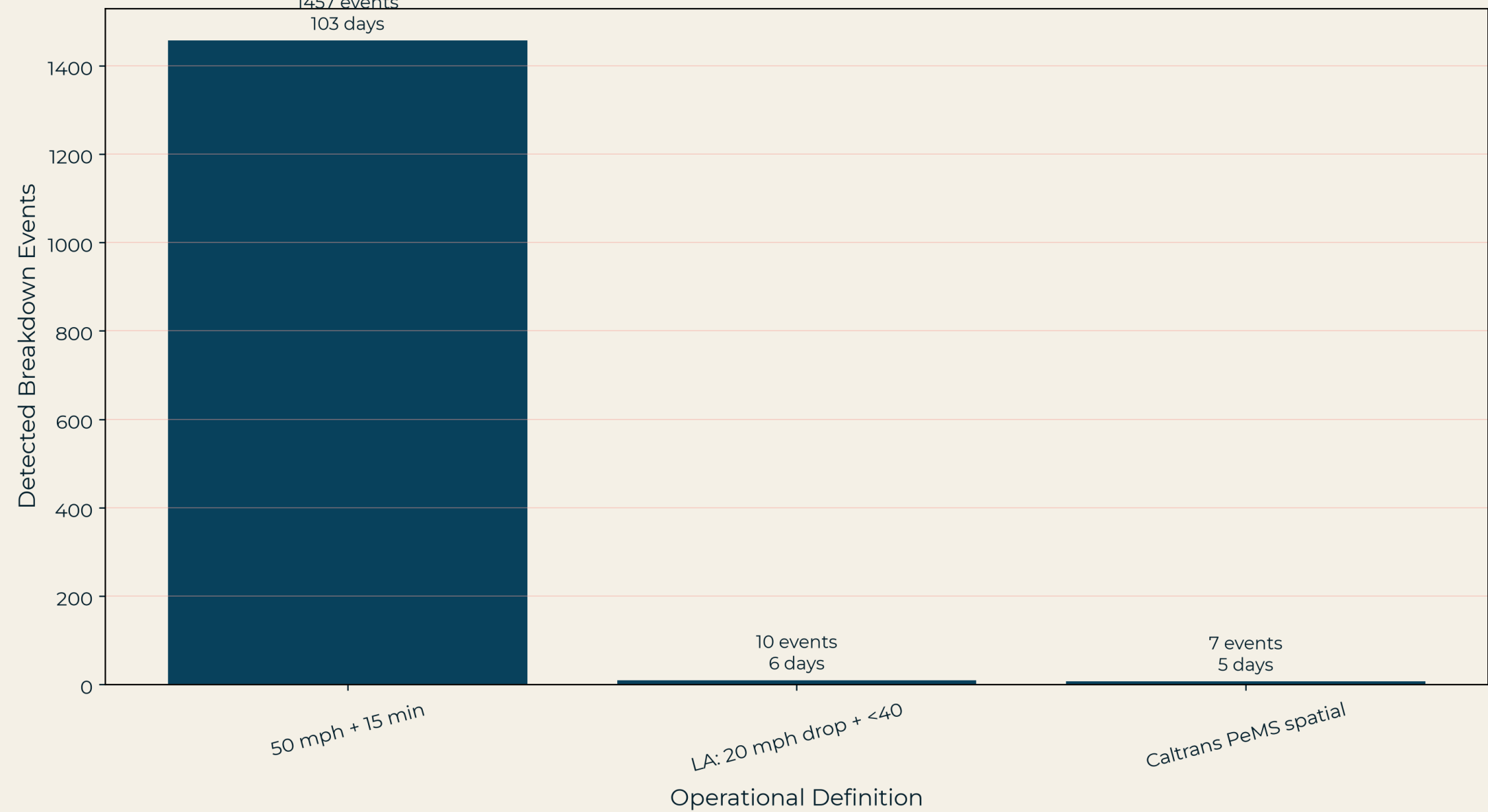


## Does a Higher Coverage Threshold Break the Corridor?

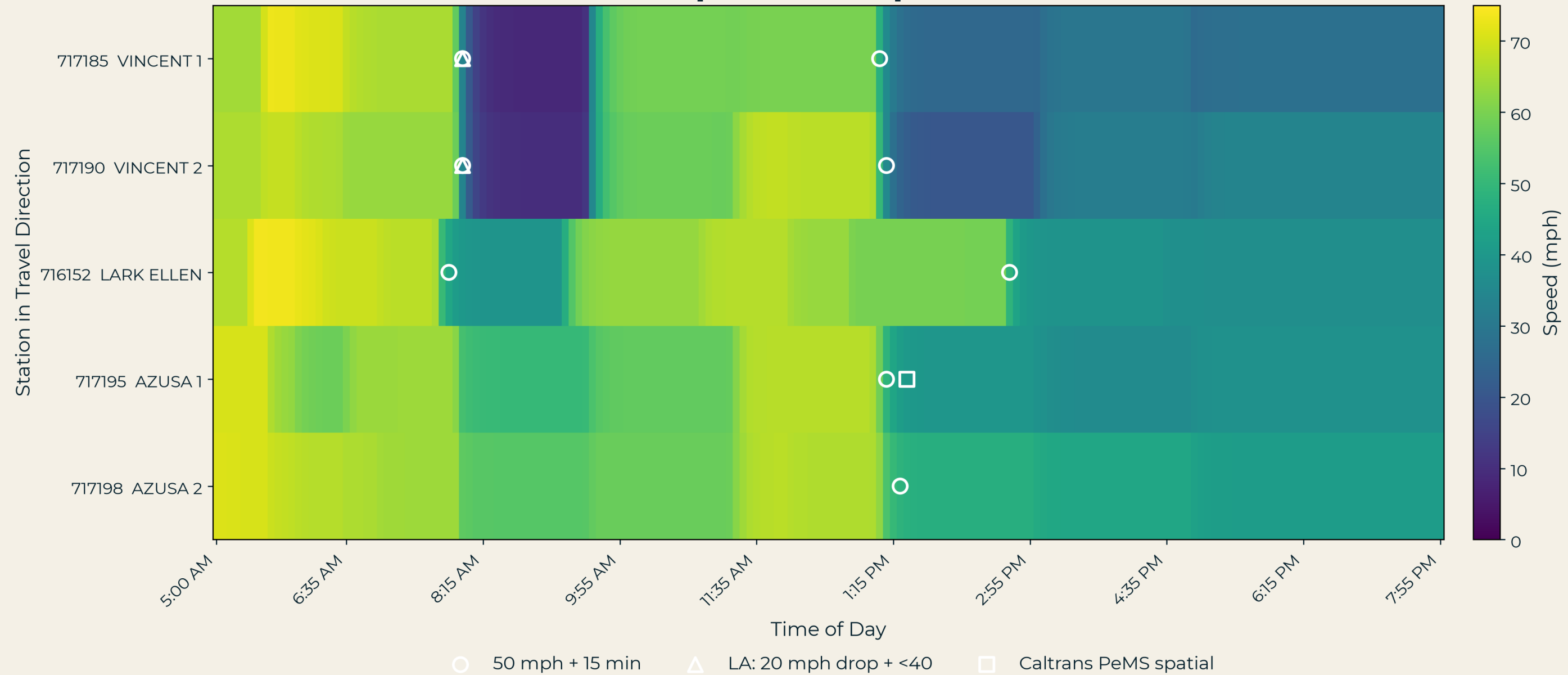


Weakest-link retention matters for propagation analysis: one poor middle station can break the spatial chain.

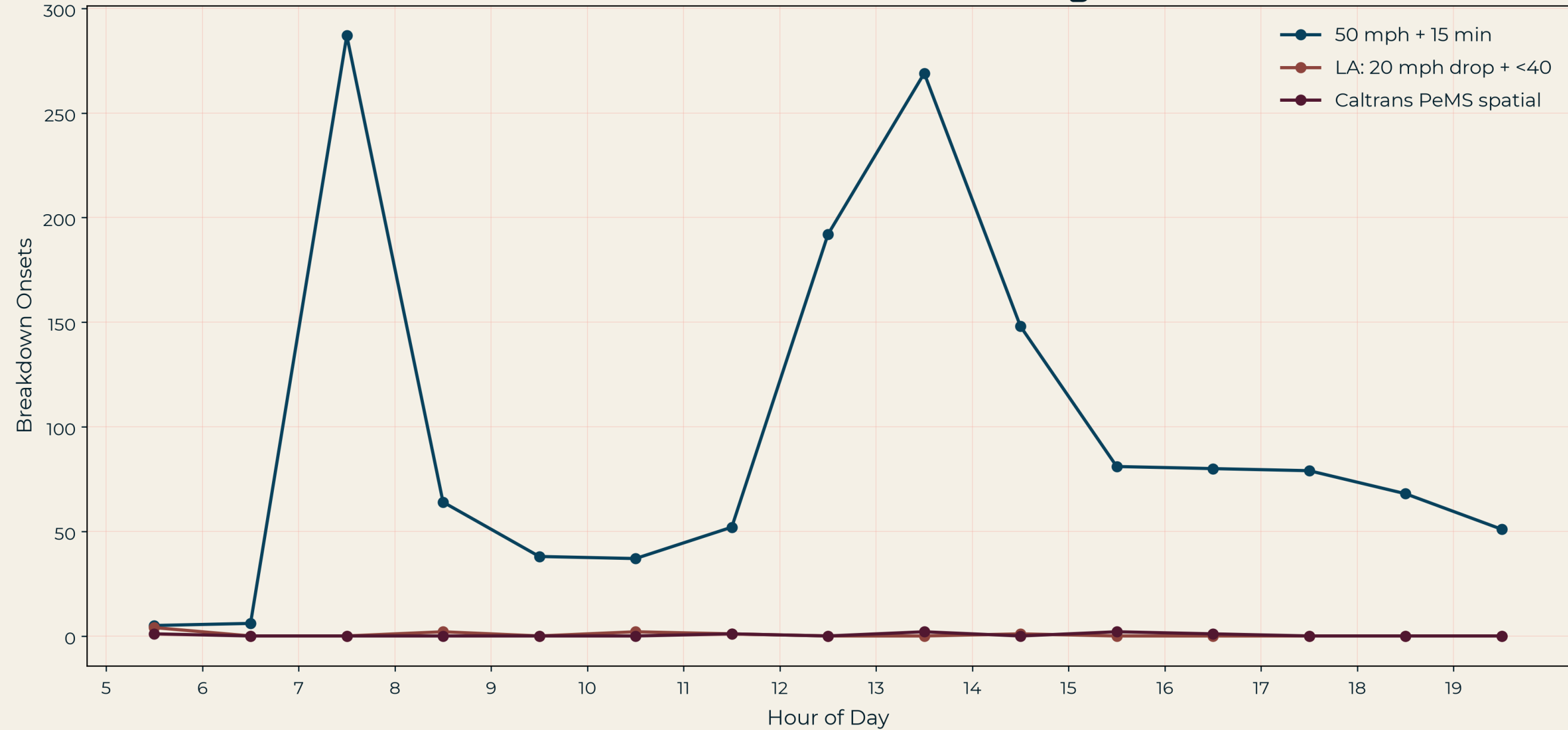
# Different Breakdown Definitions Find Different Event Sets



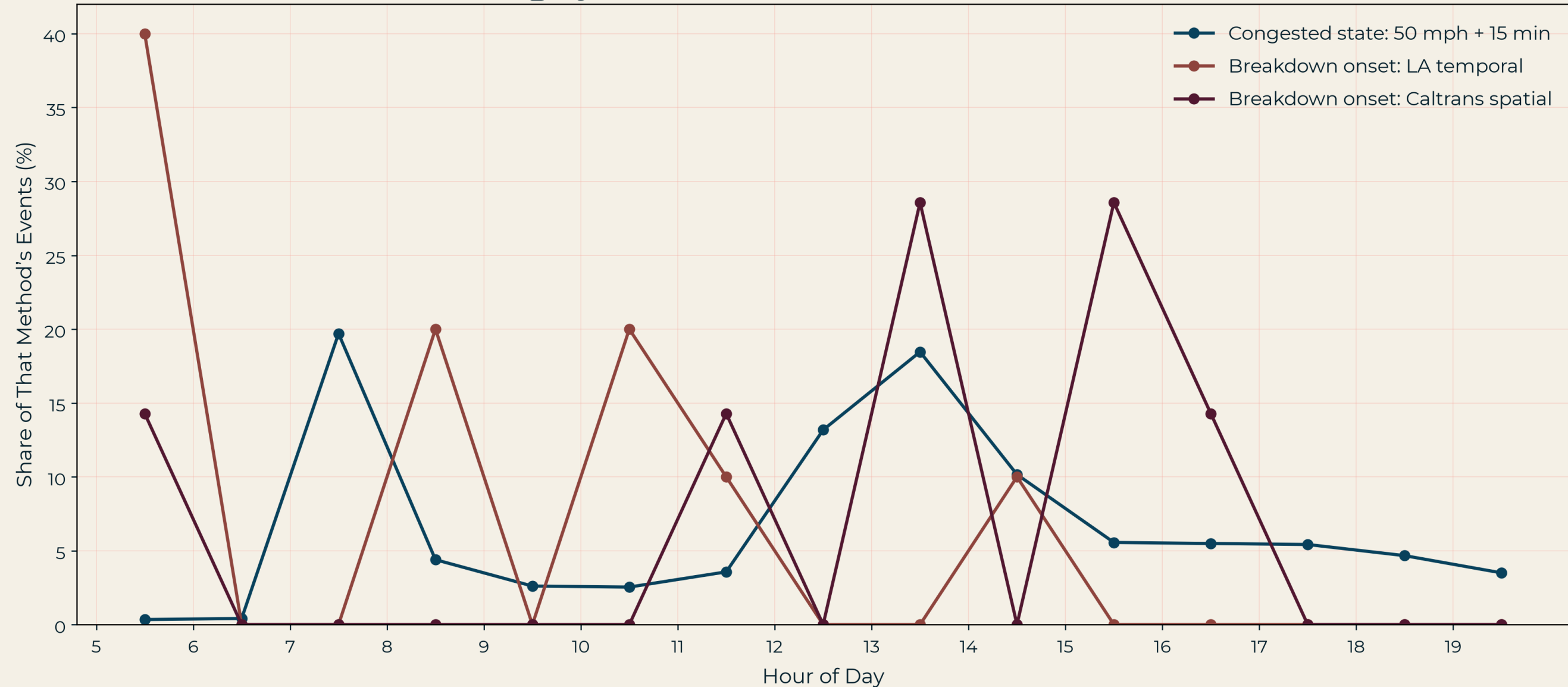
# I-10 East Speed Heatmap — 2026-02-27



## When Do Detected Breakdowns Begin?

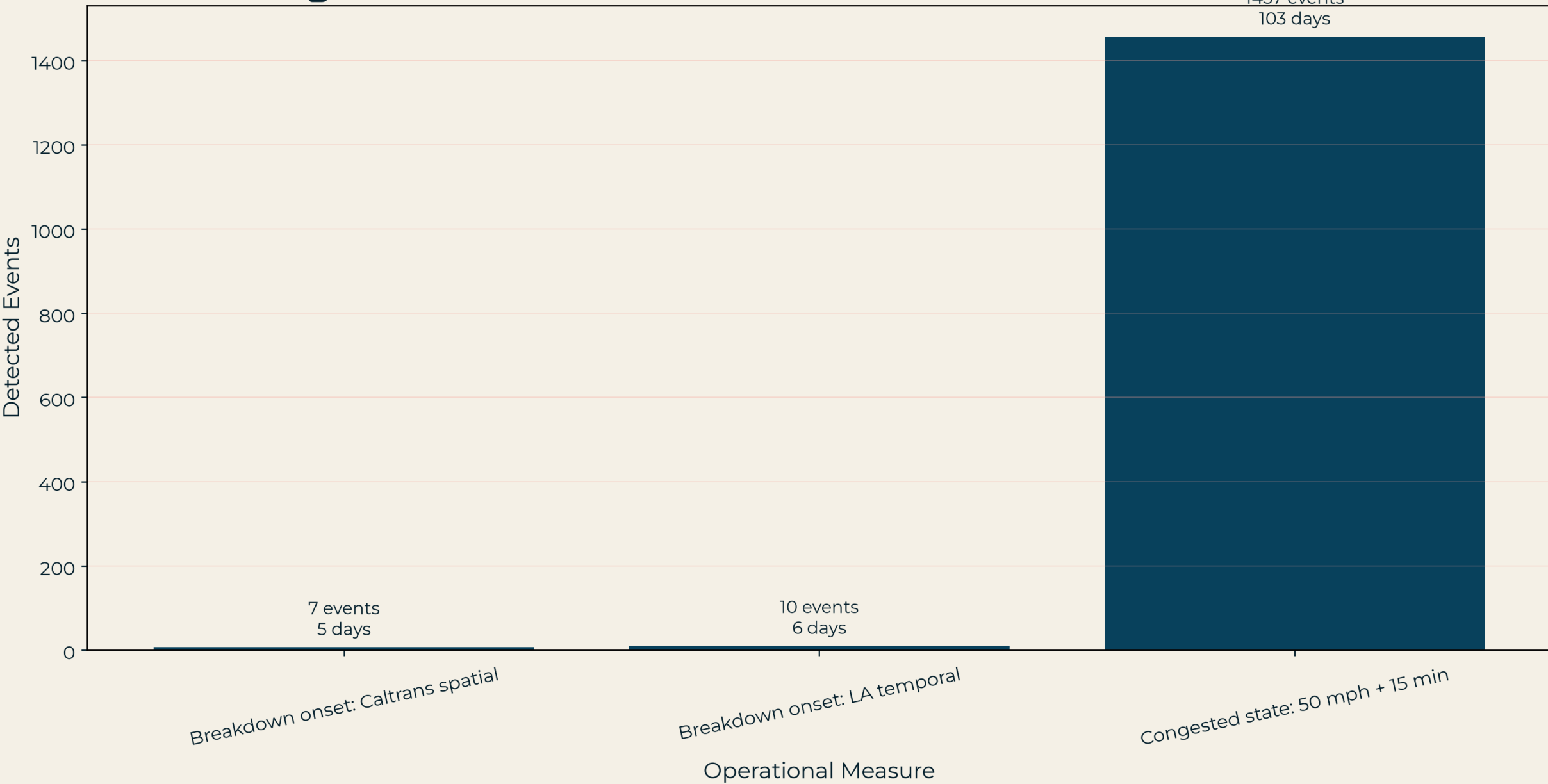


## Event Timing by Method — Normalized Within Each Definition

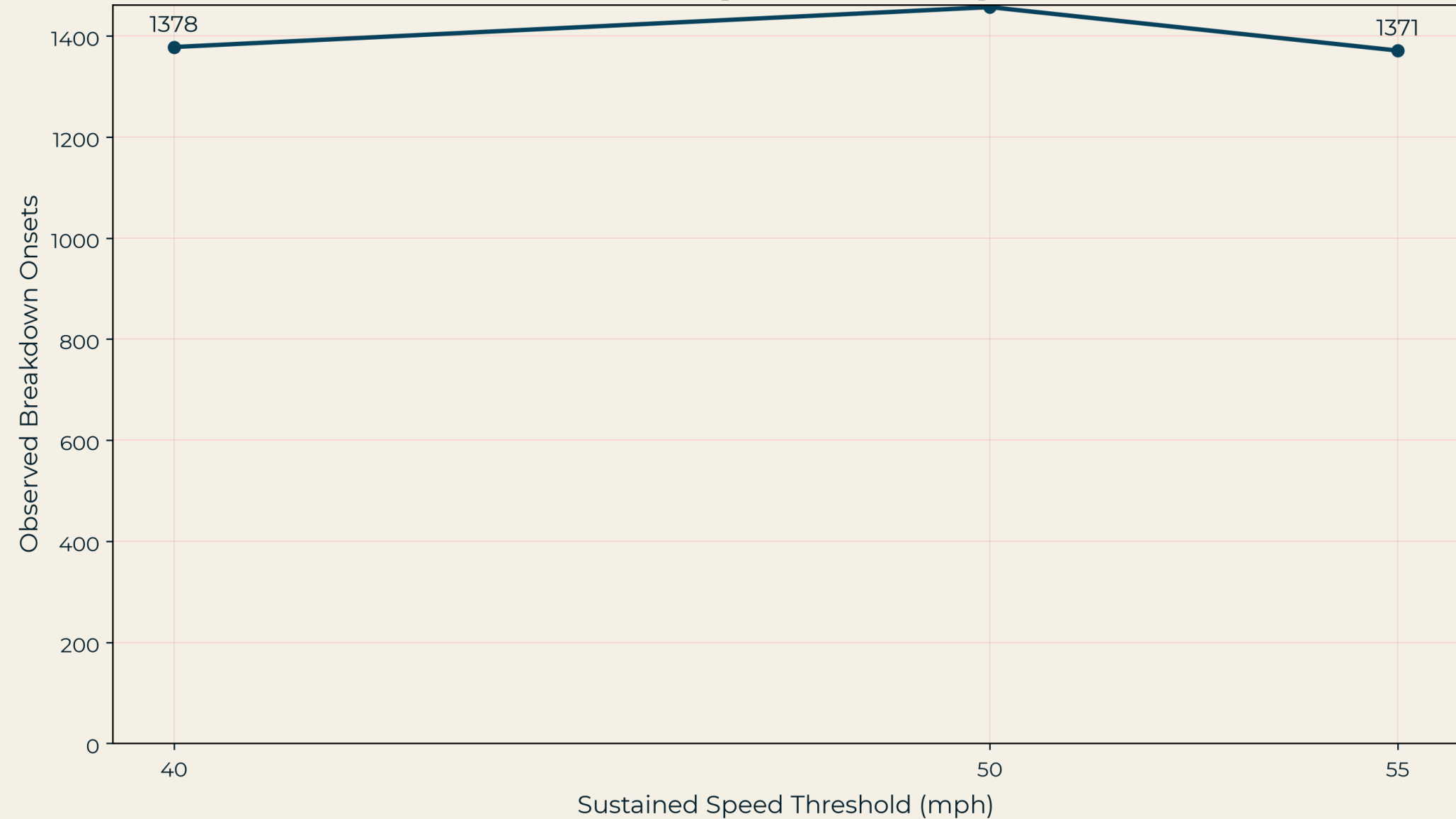


Normalization preserves timing patterns without allowing the high-frequency congestion-state method to flatten rare onset methods.

# Congested State and Breakdown Onset Are Different Measurements

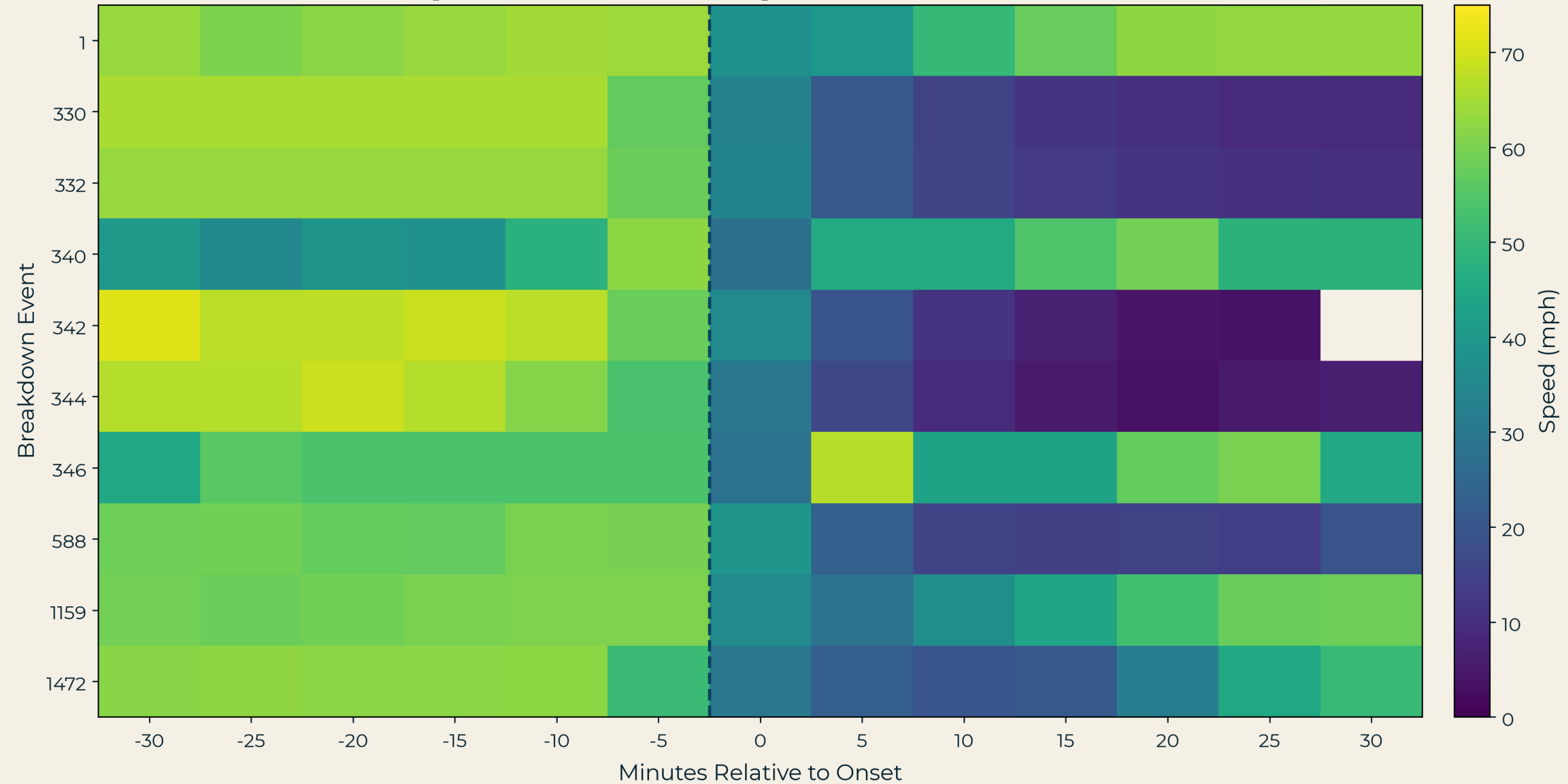


# Breakdown Counts Depend on the Speed Threshold

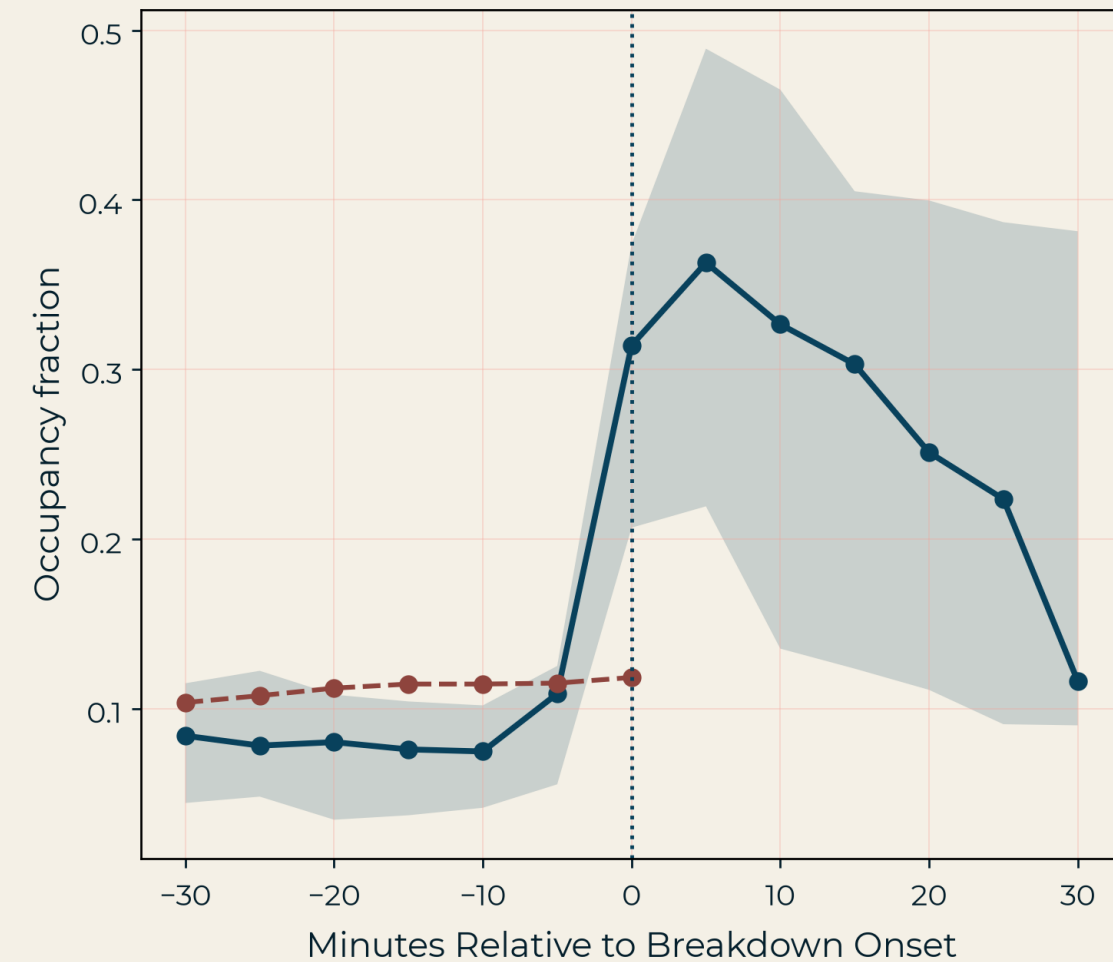
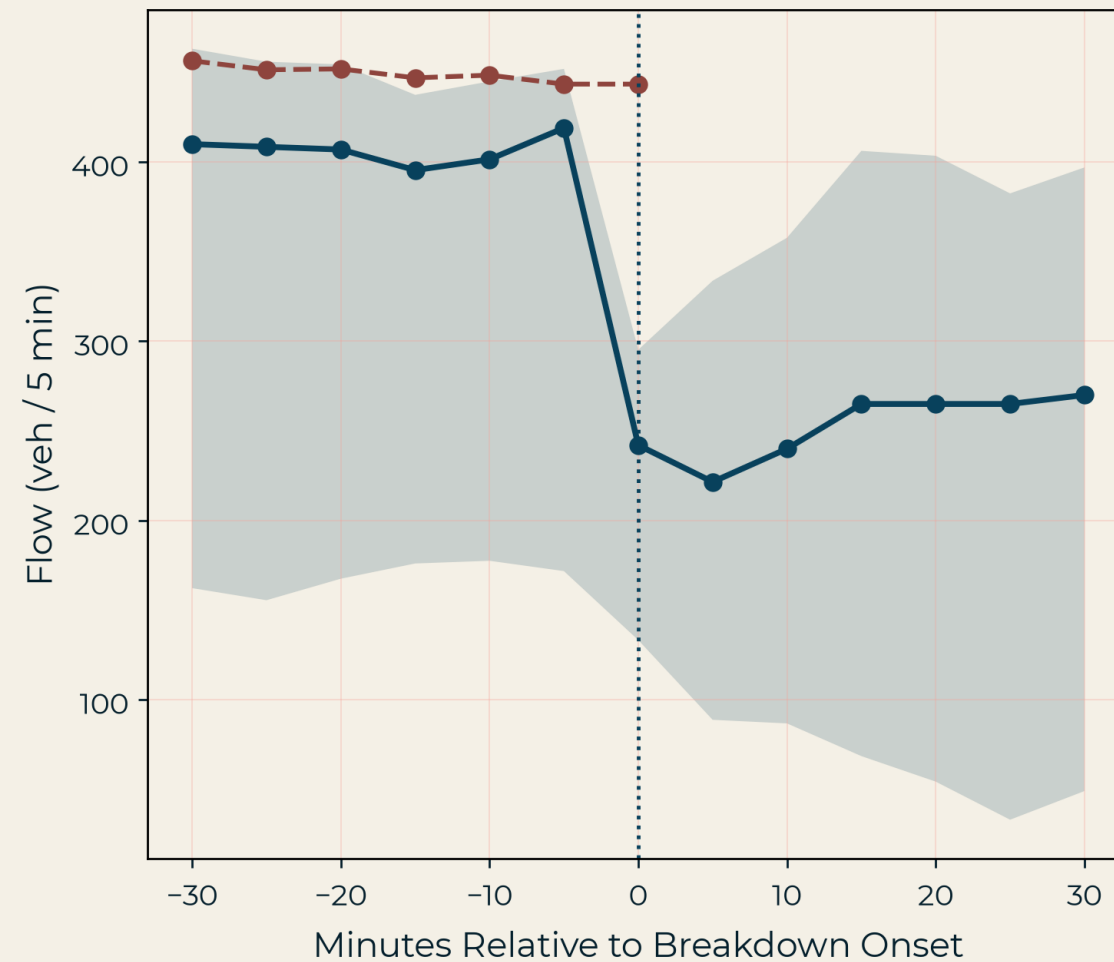
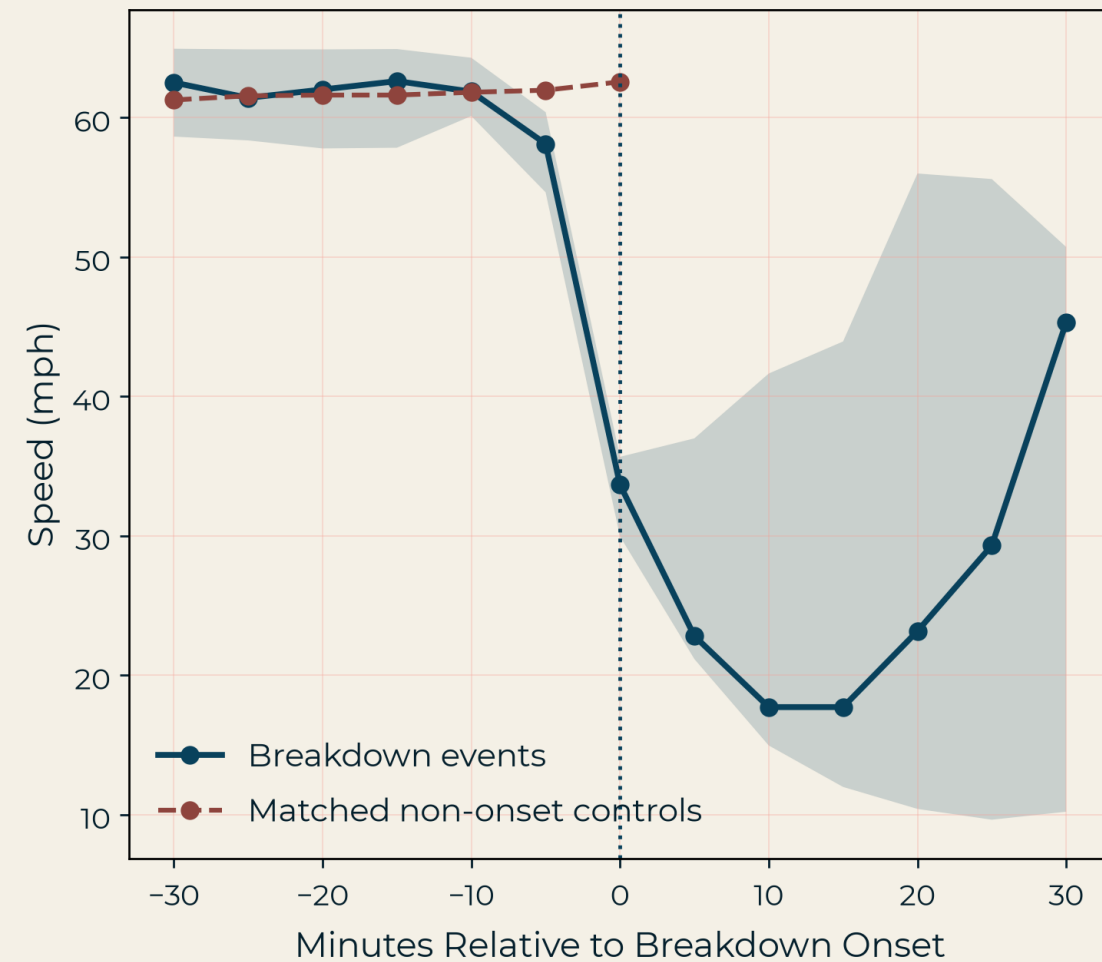


Each threshold requires at least 15 consecutive minutes below the selected speed.

# Ten Abrupt Breakdowns — Speed Before and After Onset

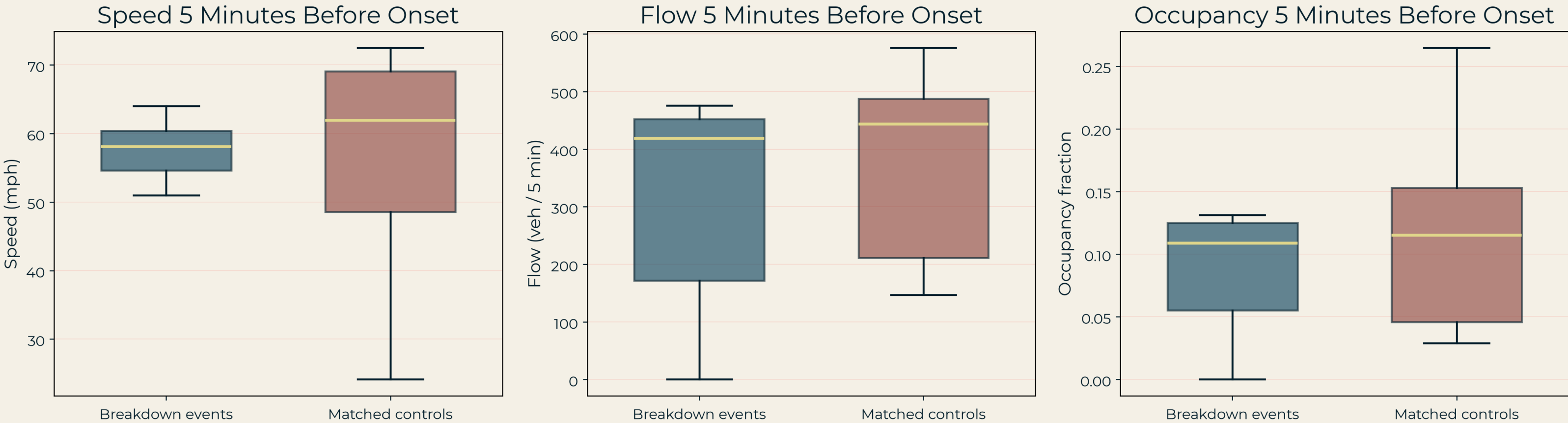


# What Happens Around an Abrupt Traffic Breakdown?



Primary onset definition: LA temporal rule ( $\geq 20$  mph five-minute drop with resulting speed  $< 40$  mph). Shaded bands are event IQRs.

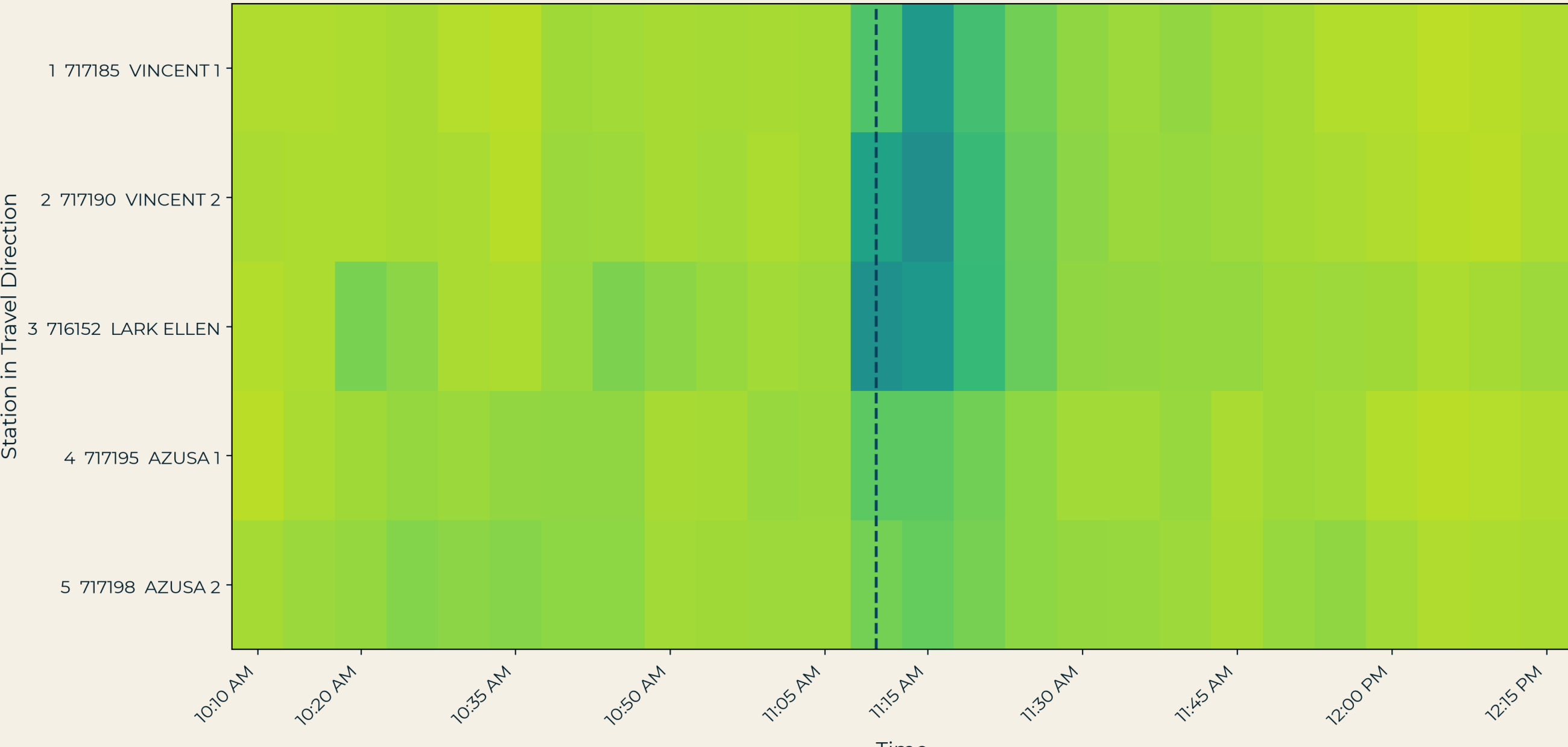
# Are Conditions Already Different Before the Speed Collapse?



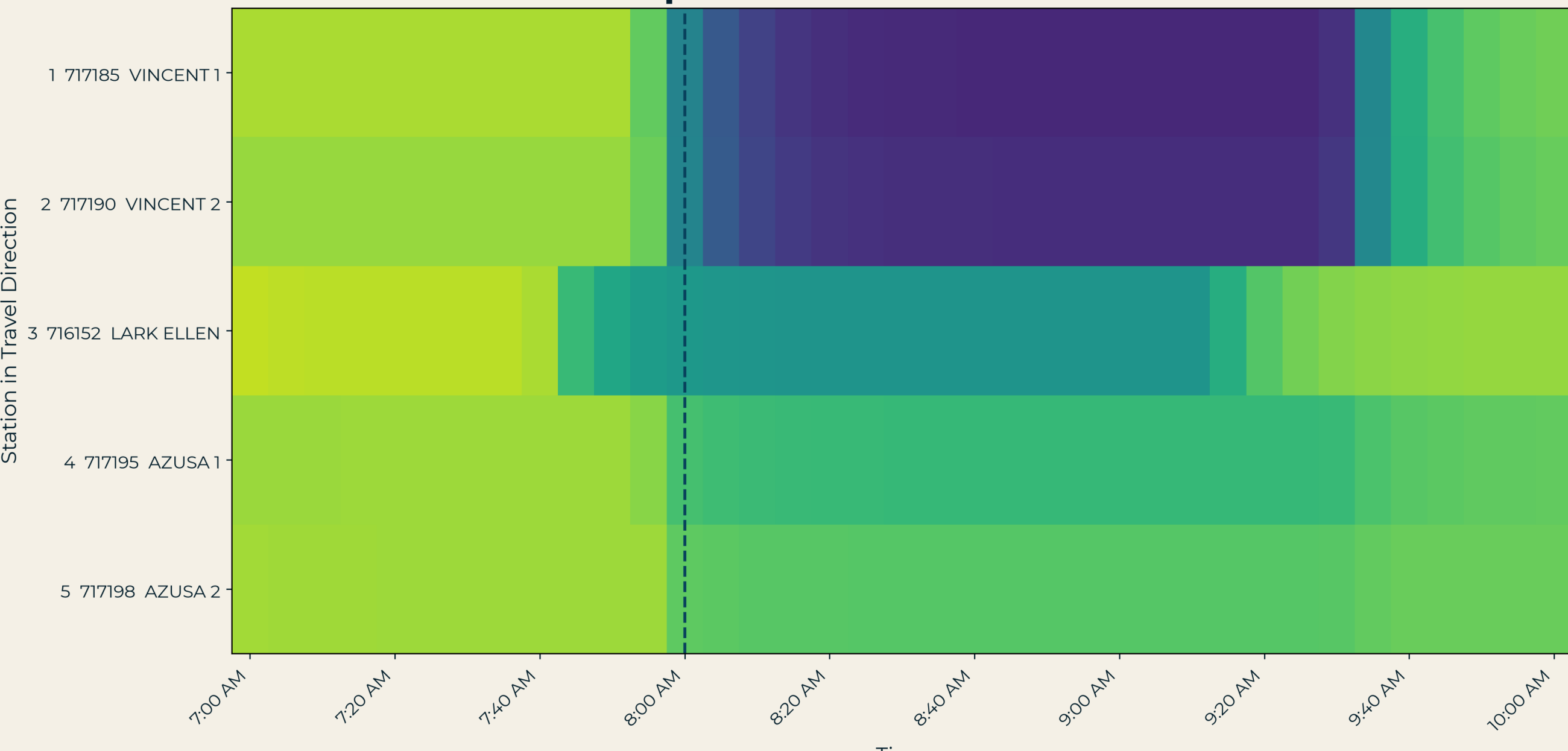
Controls match station, weekday/weekend class, and half-hour clock bin; controls within 60 minutes of an LA onset are excluded.

Raw Speed Evolution Around Corridor-Level Breakdown Episodes

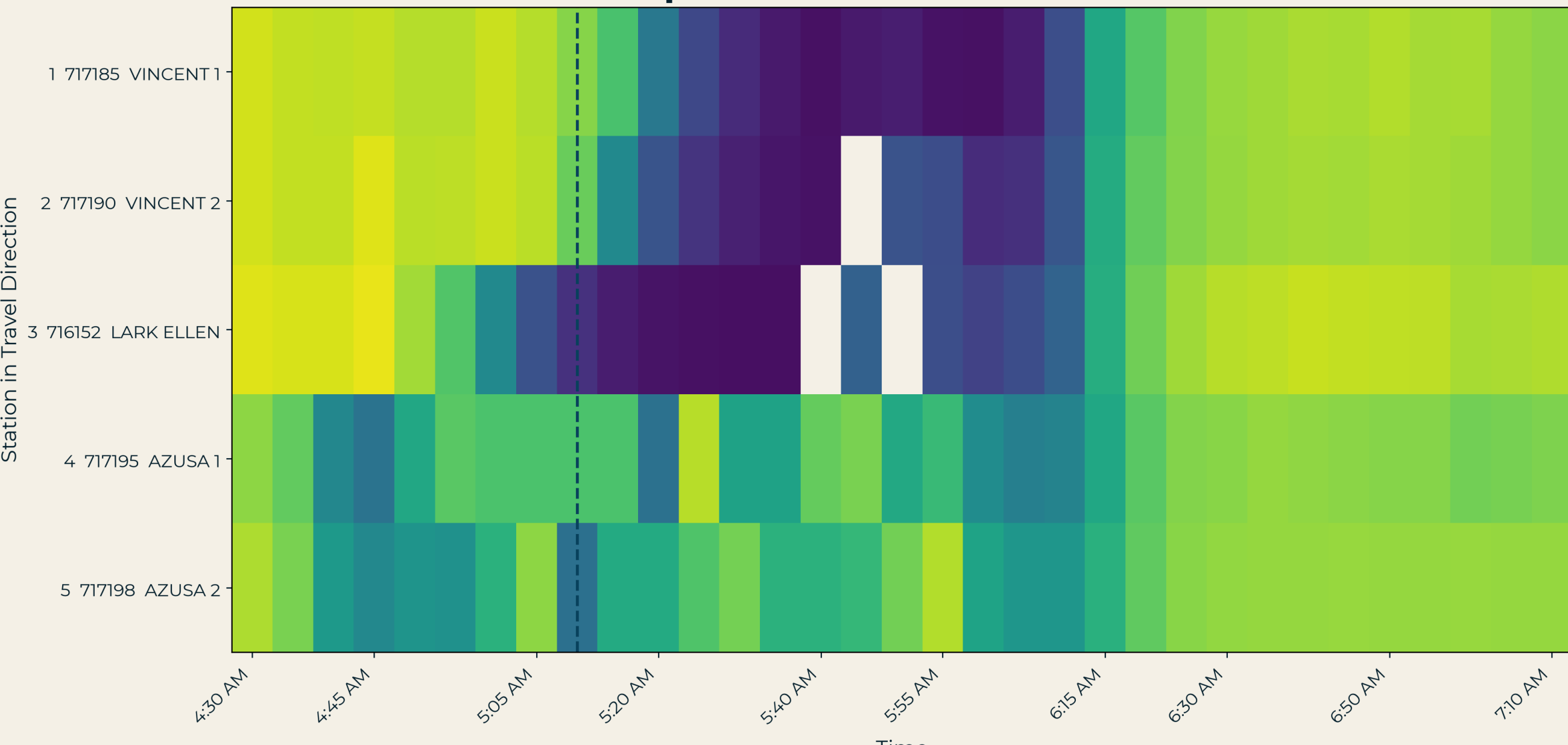
Episode 1 — 2026-02-01 11:10 AM



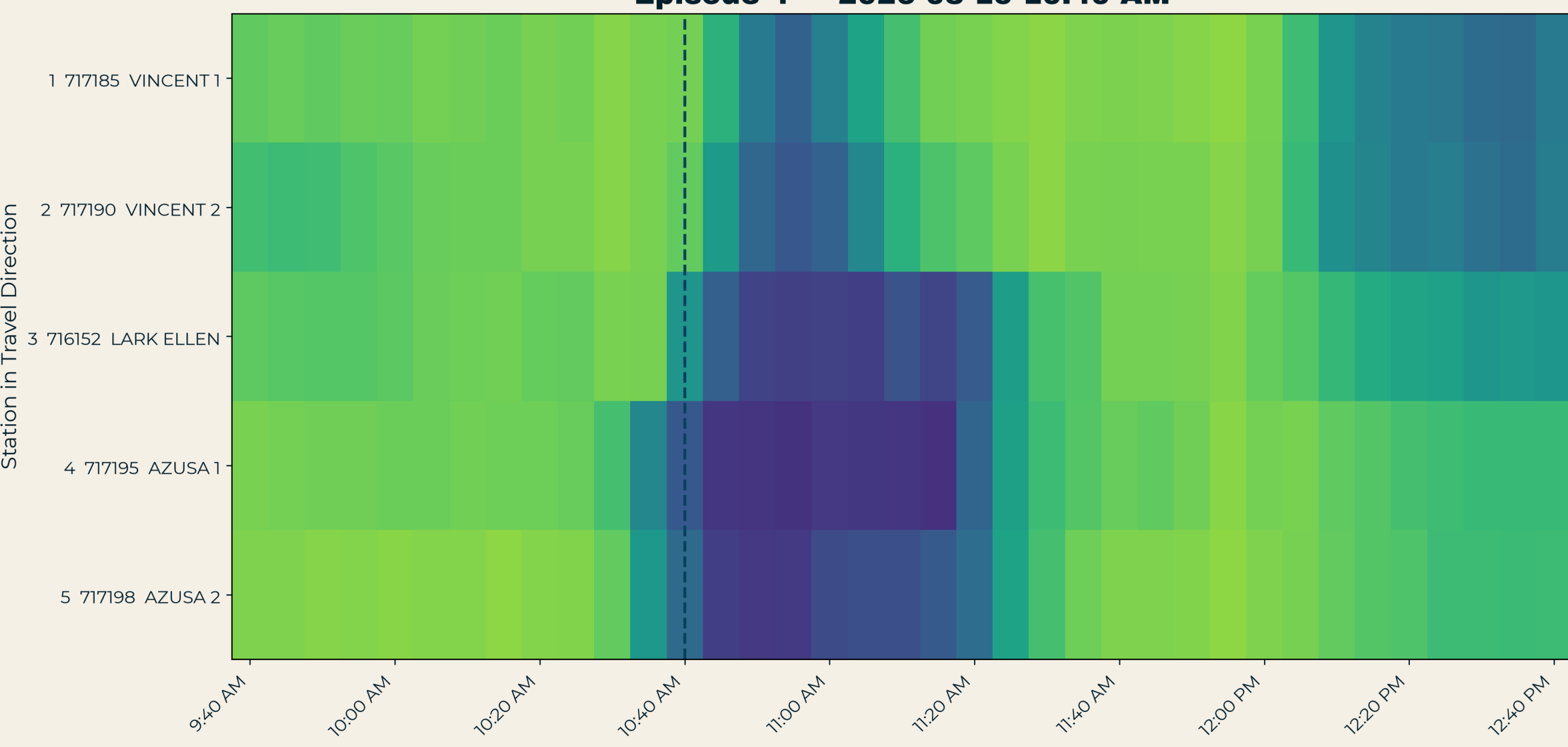
Episode 2 — 2026-02-27 8:00 AM



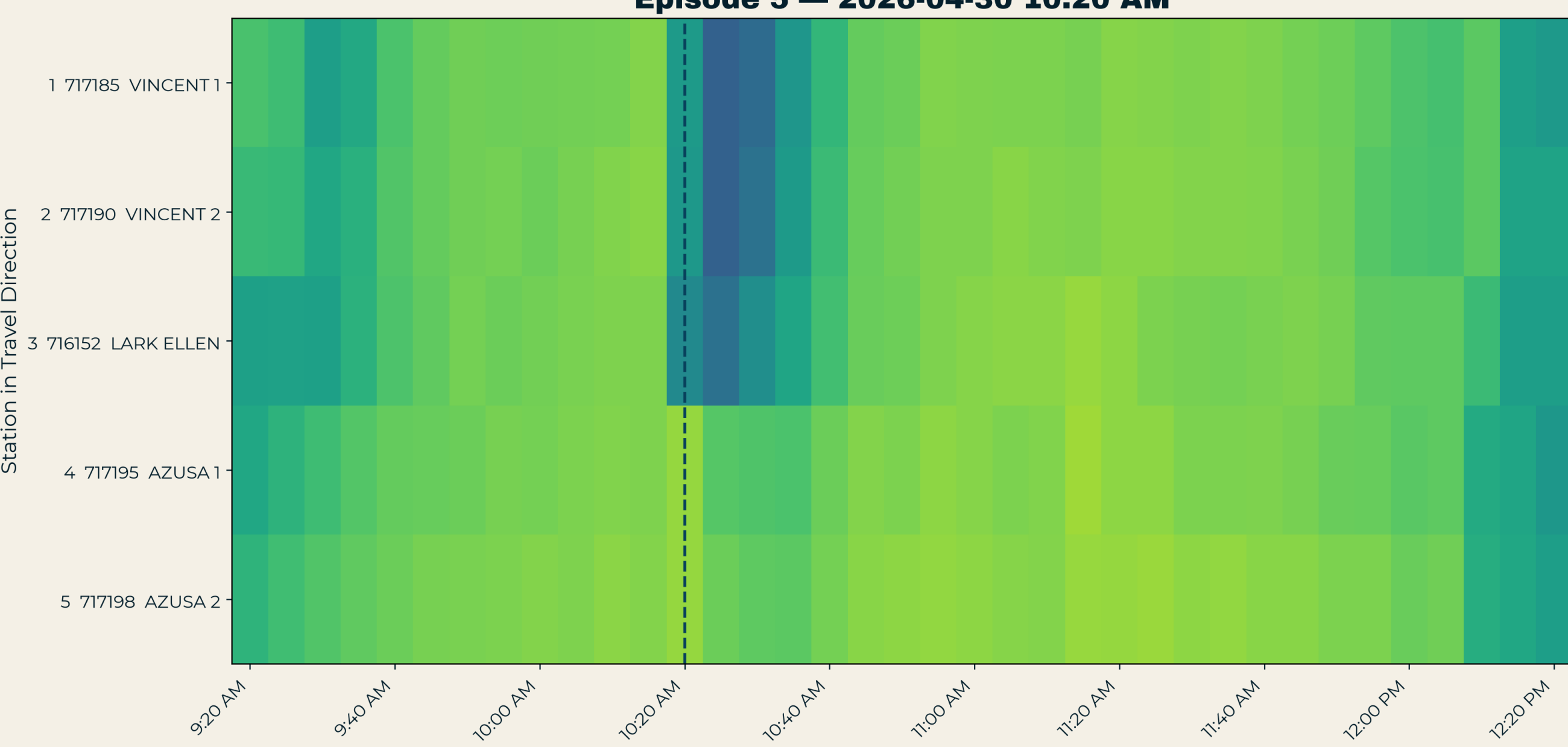
Episode 3 — 2026-03-02 5:10 AM



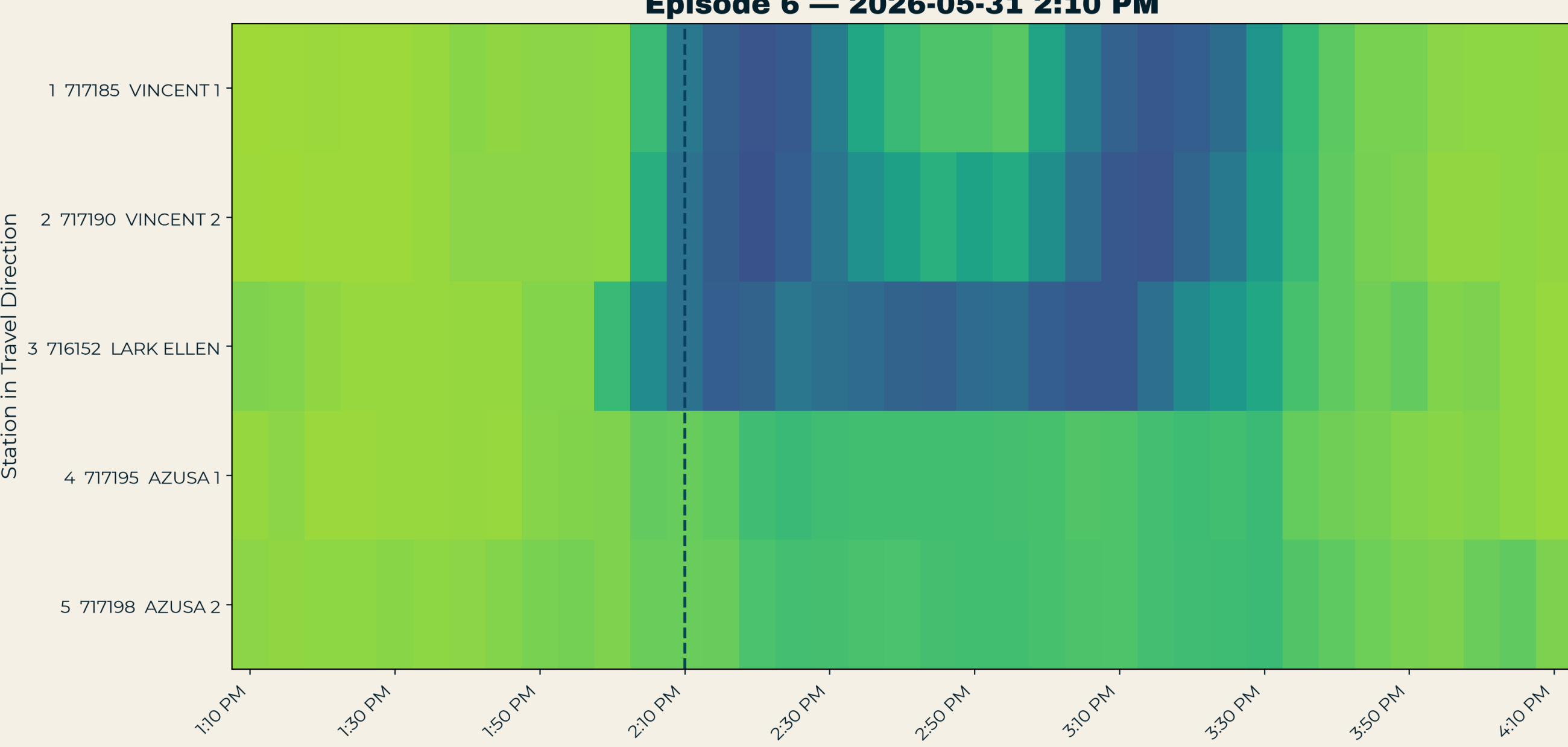
Episode 4 — 2026-03-19 10:40 AM



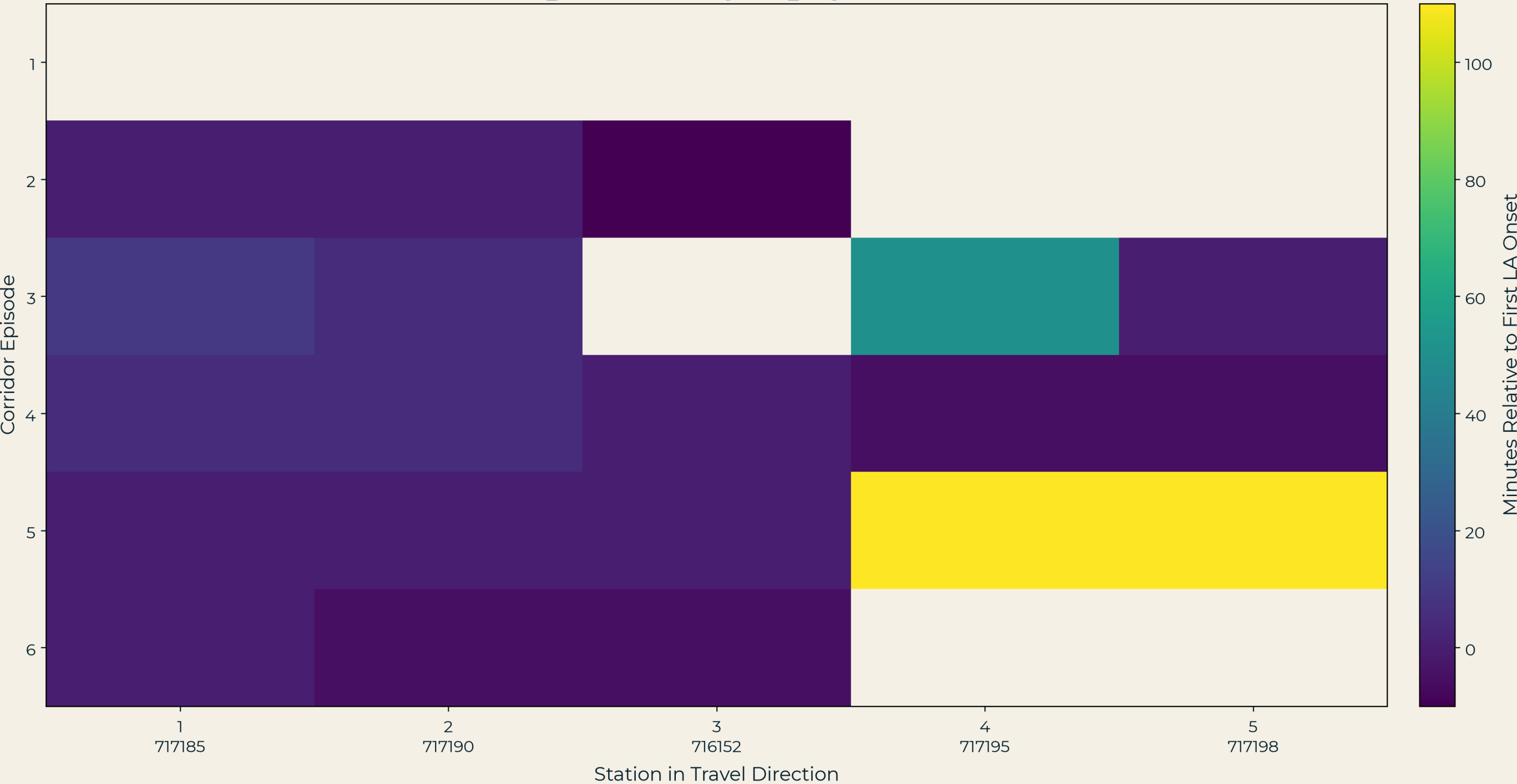
Episode 5 — 2026-04-30 10:20 AM



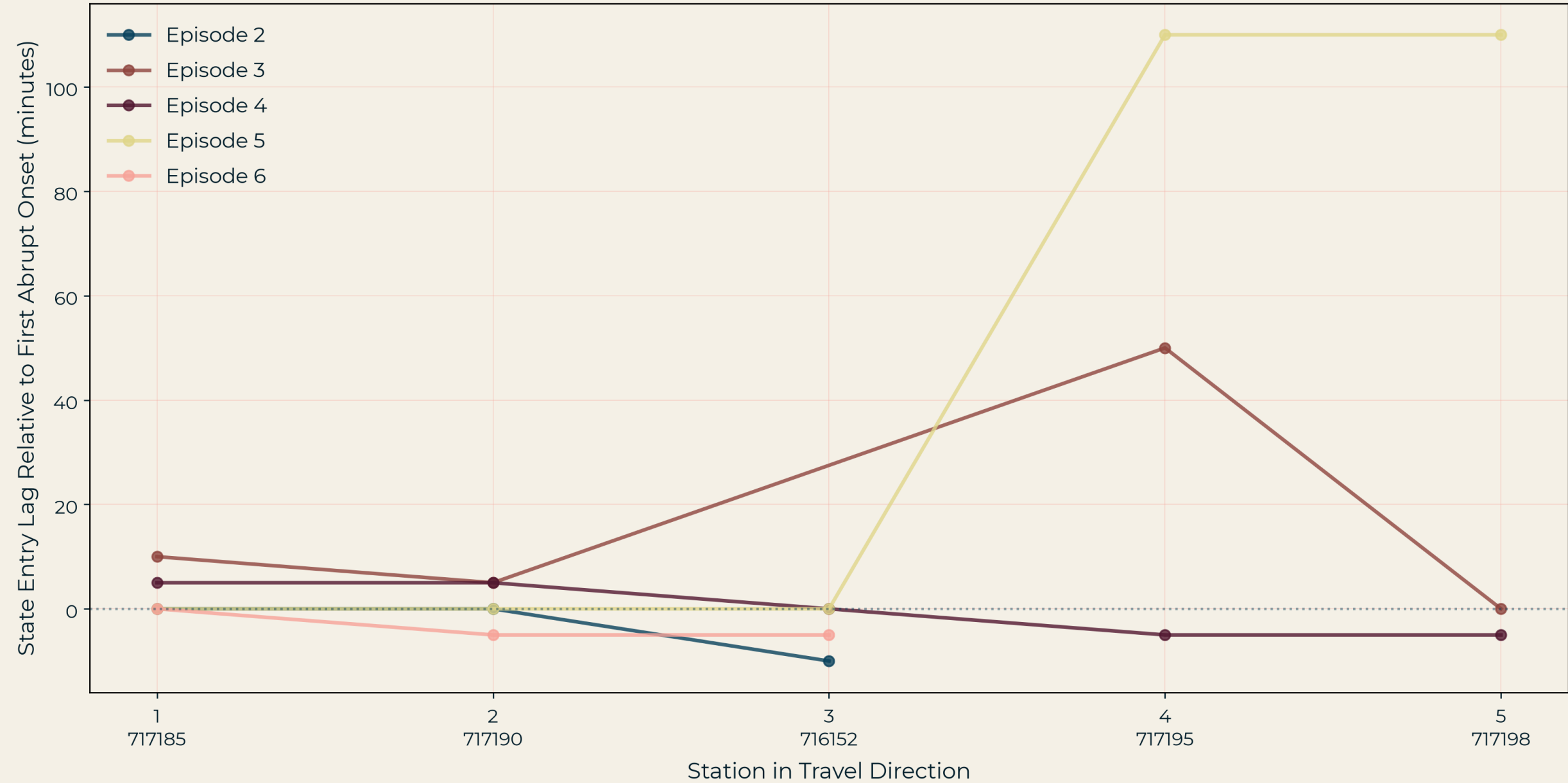
Episode 6 — 2026-05-31 2:10 PM



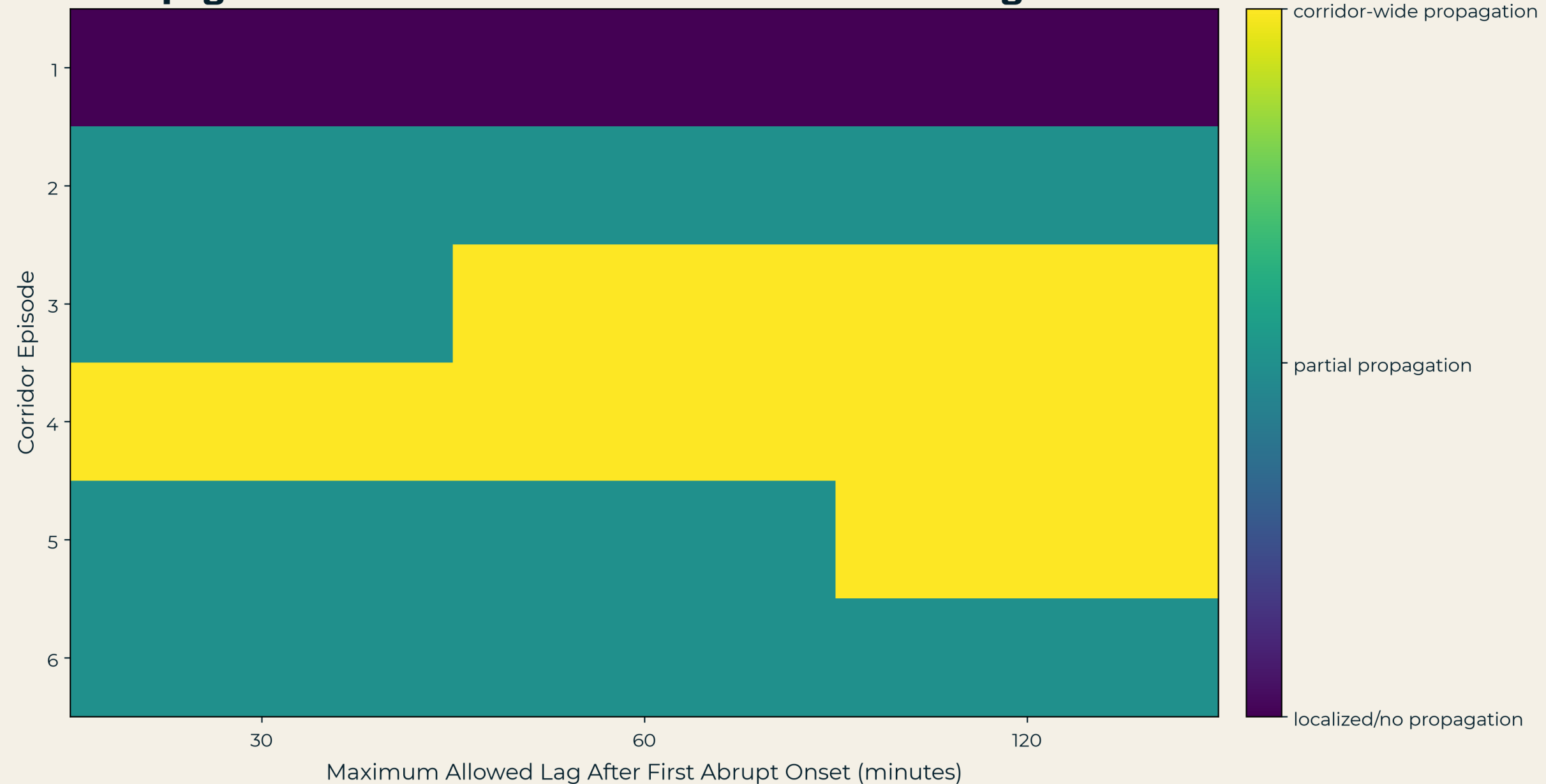
**Sustained-Congestion Entry Lag by Corridor Episode**



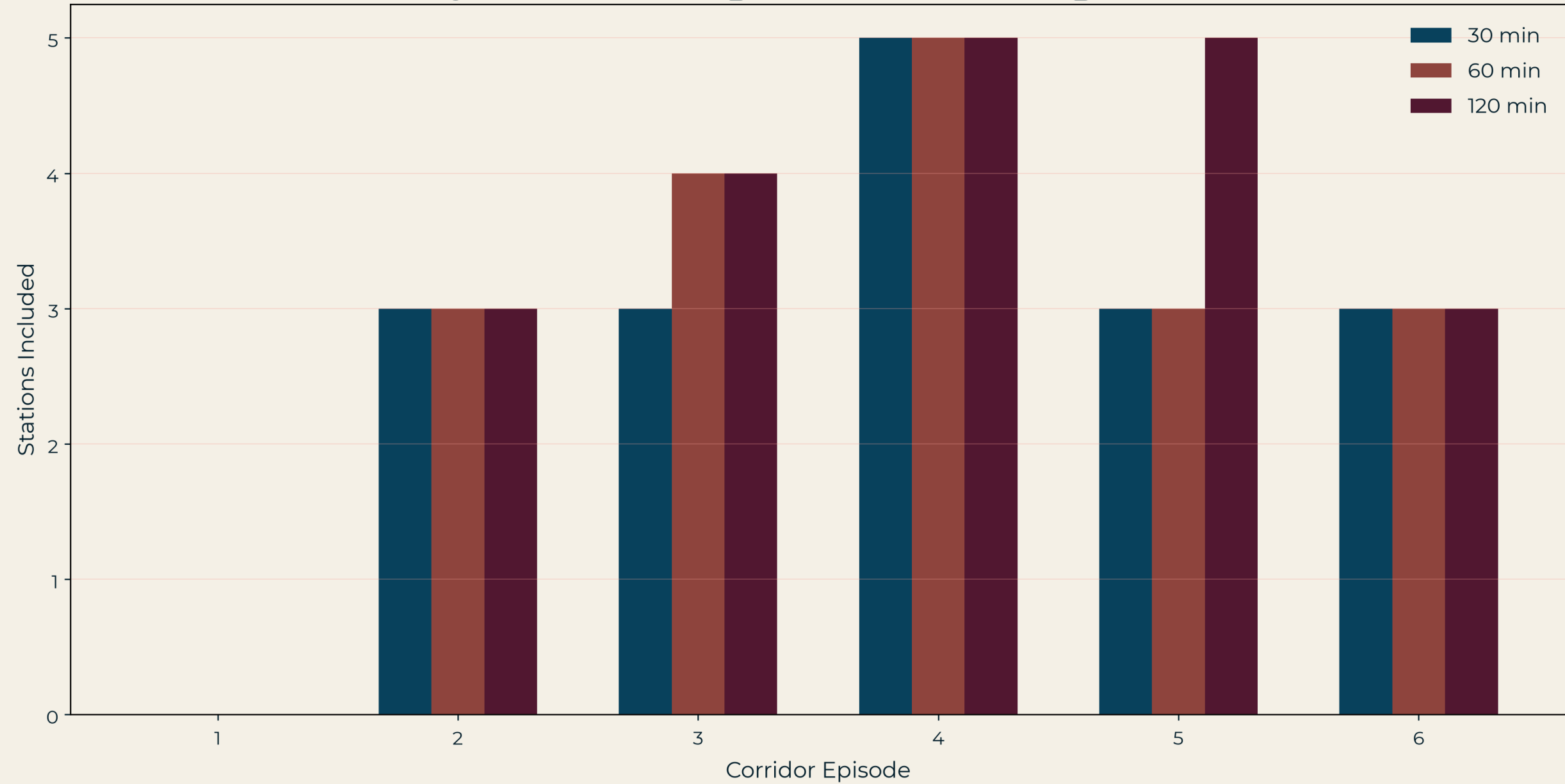
# How Quickly Does Sustained Congestion Appear Across the Corridor?



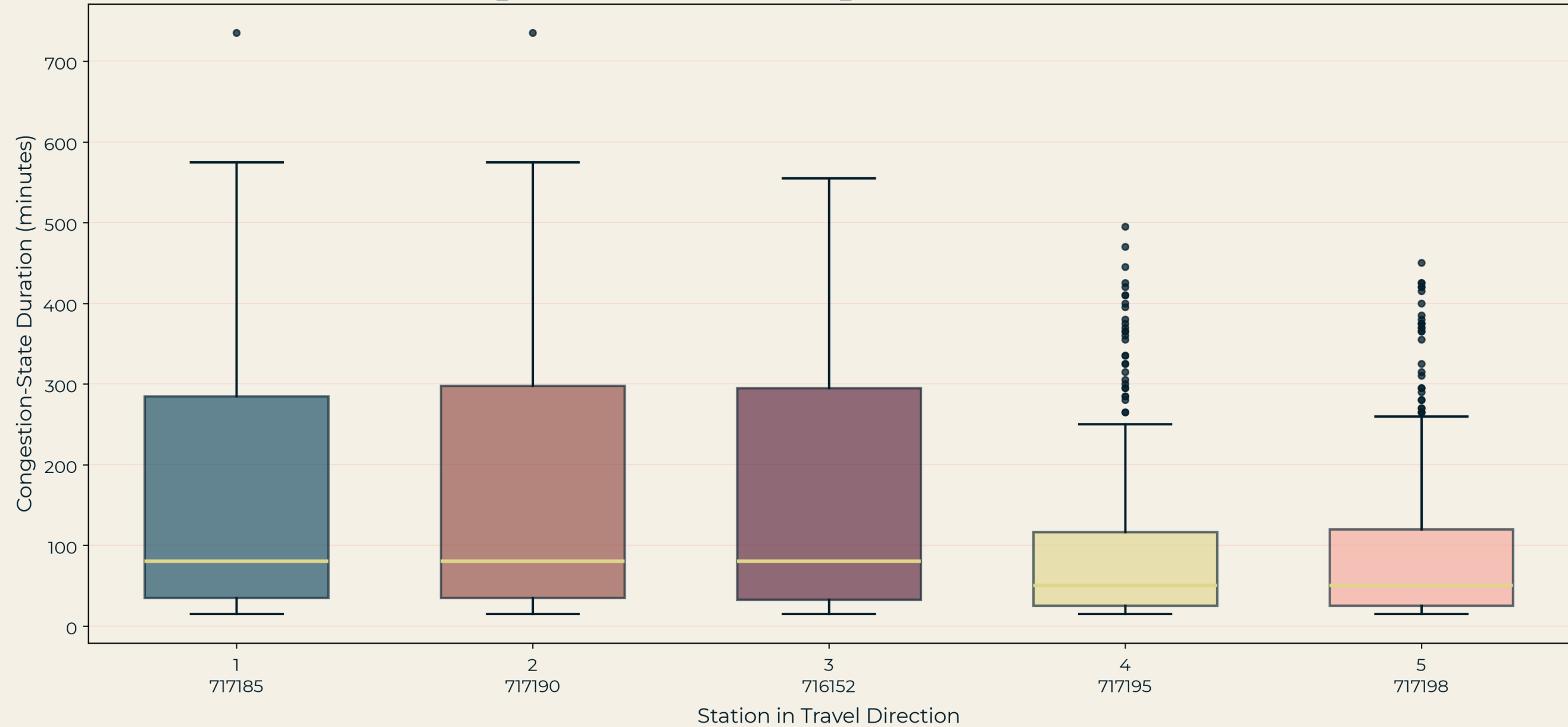
# Propagation Classification Is Tested Across Three Lag Windows



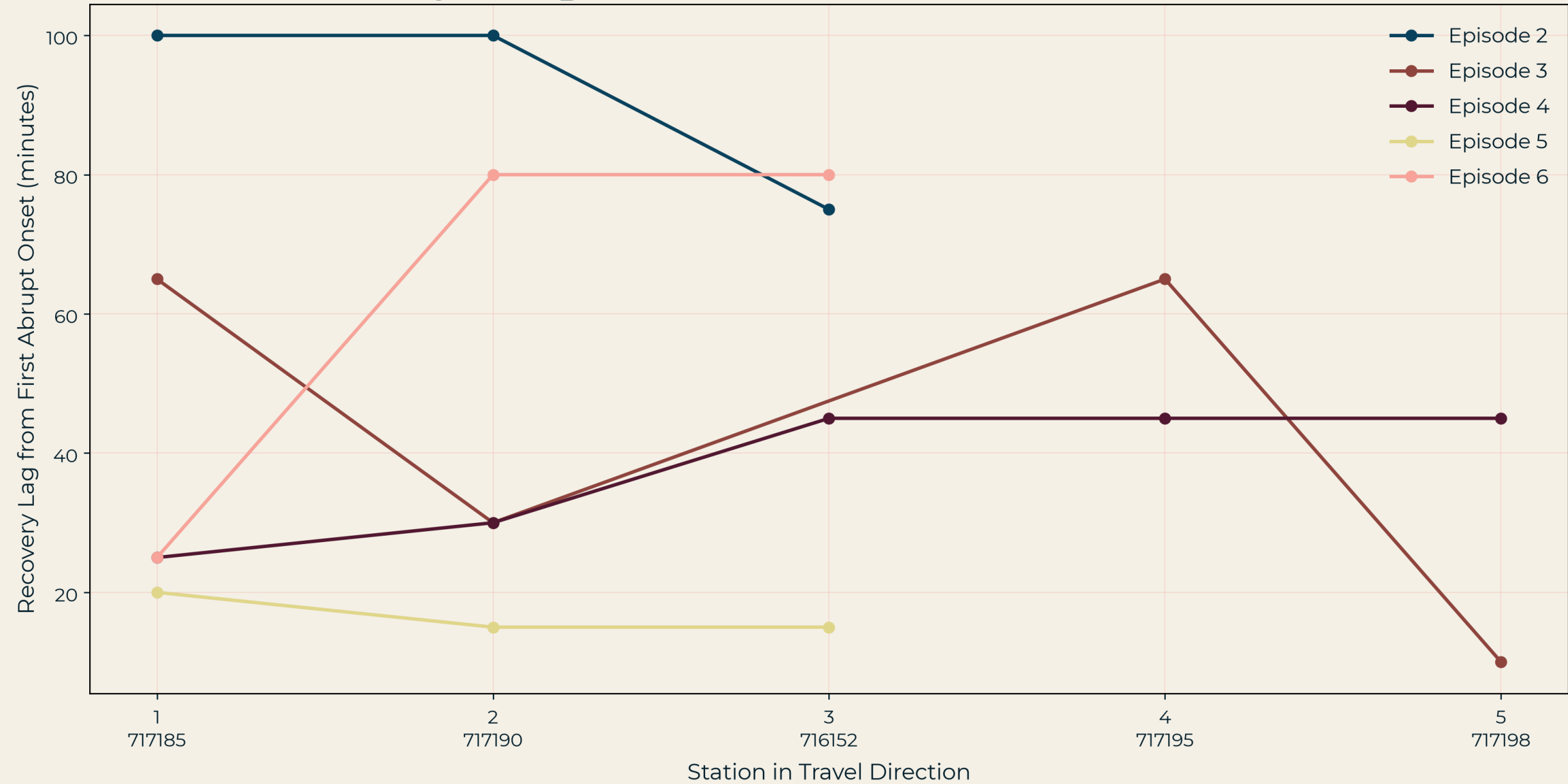
# How Many Stations Belong to the Same Propagation Episode?



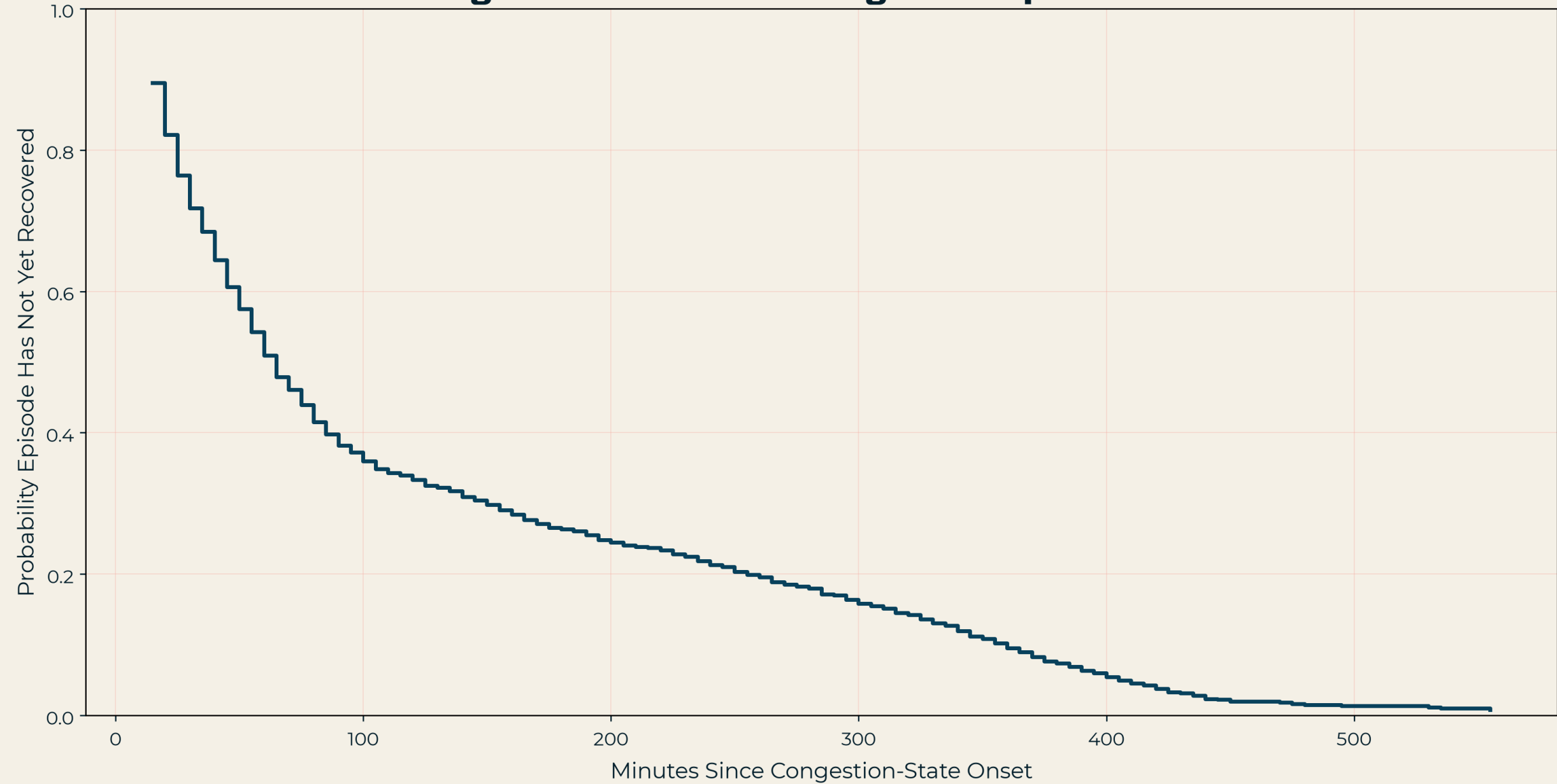
# How Long Does Sustained Congestion Last at Each Station?



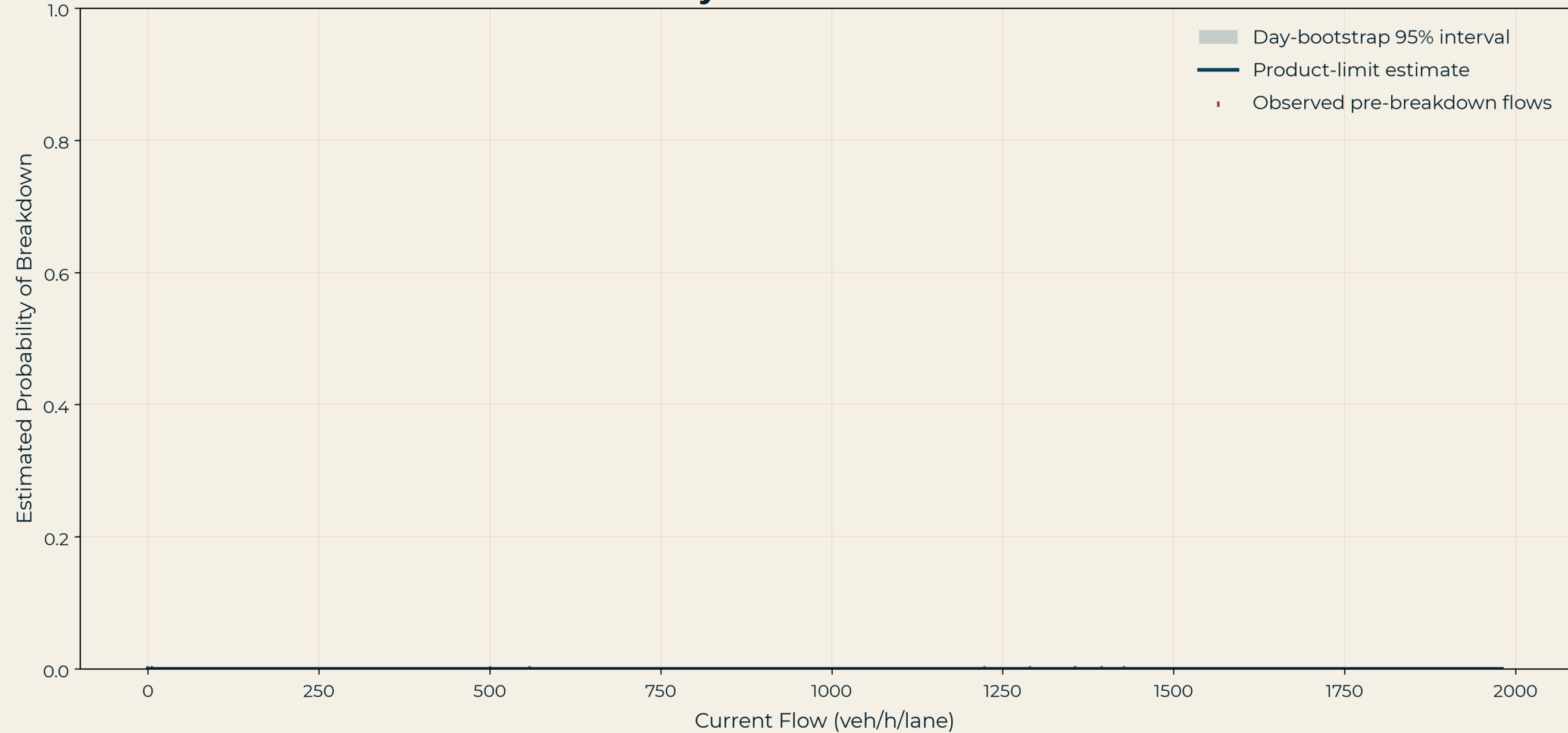
# Recovery Timing Across Stations in the Same Corridor Episode



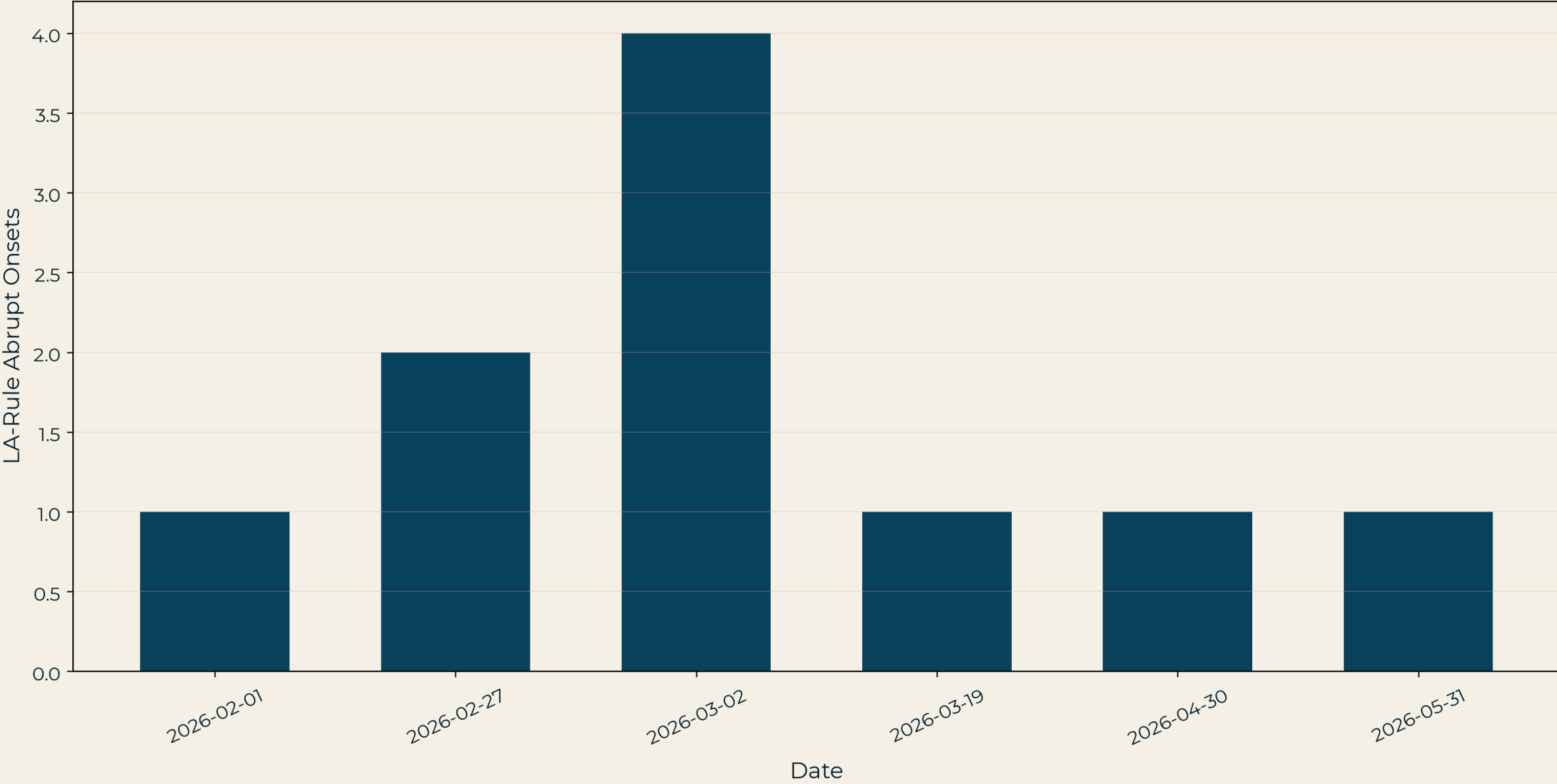
# How Long Until a Sustained Congestion Episode Recovers?



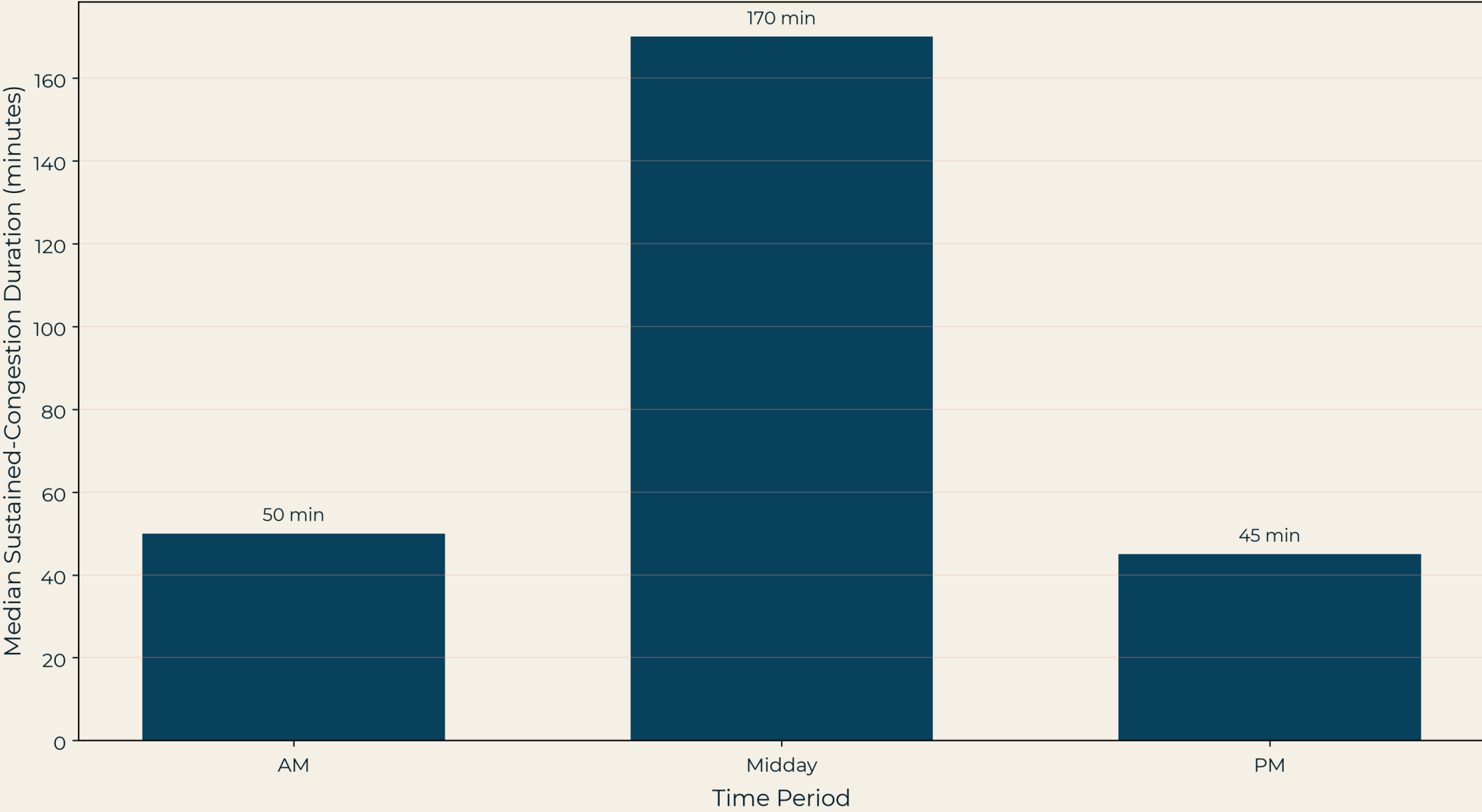
# Estimated Probability of Breakdown Increases with Flow



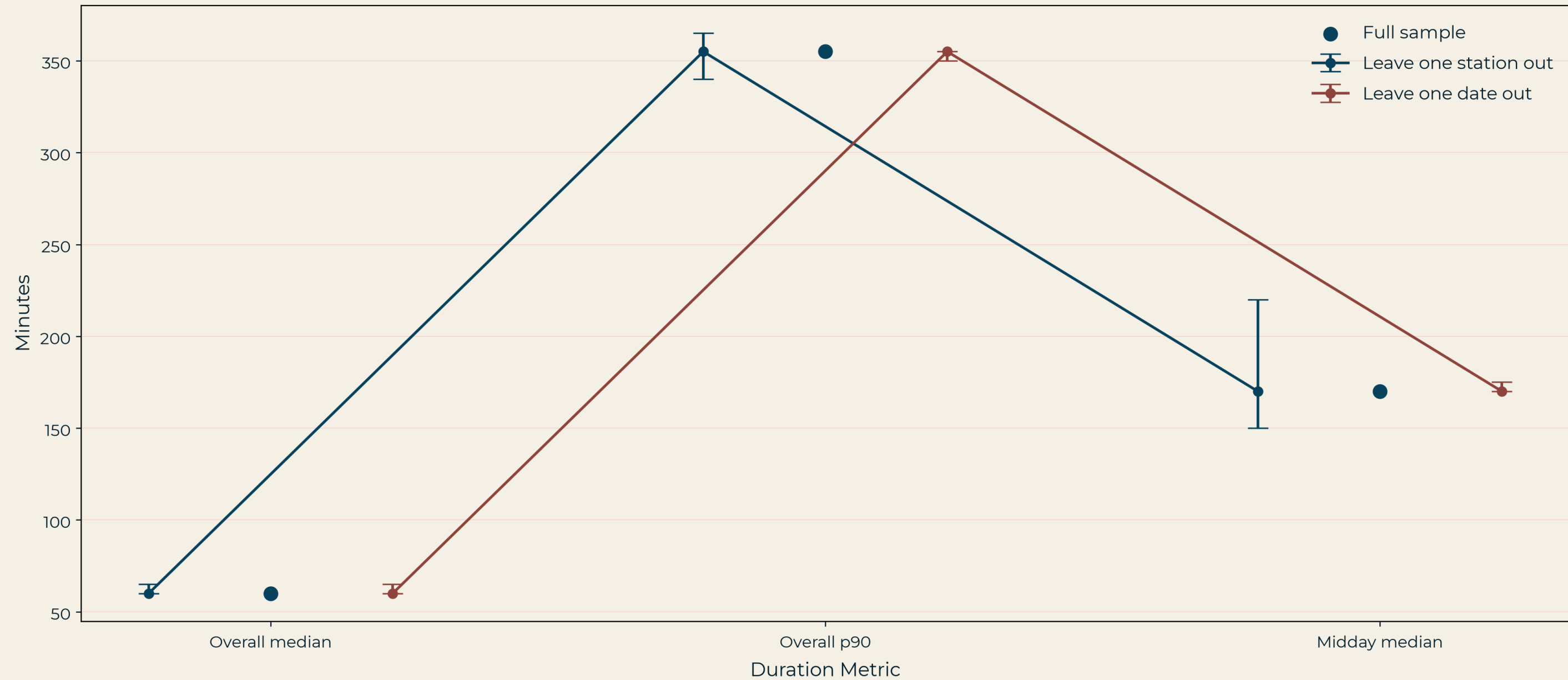
# Abrupt Breakdown Onsets Are Concentrated in a Small Number of Dates



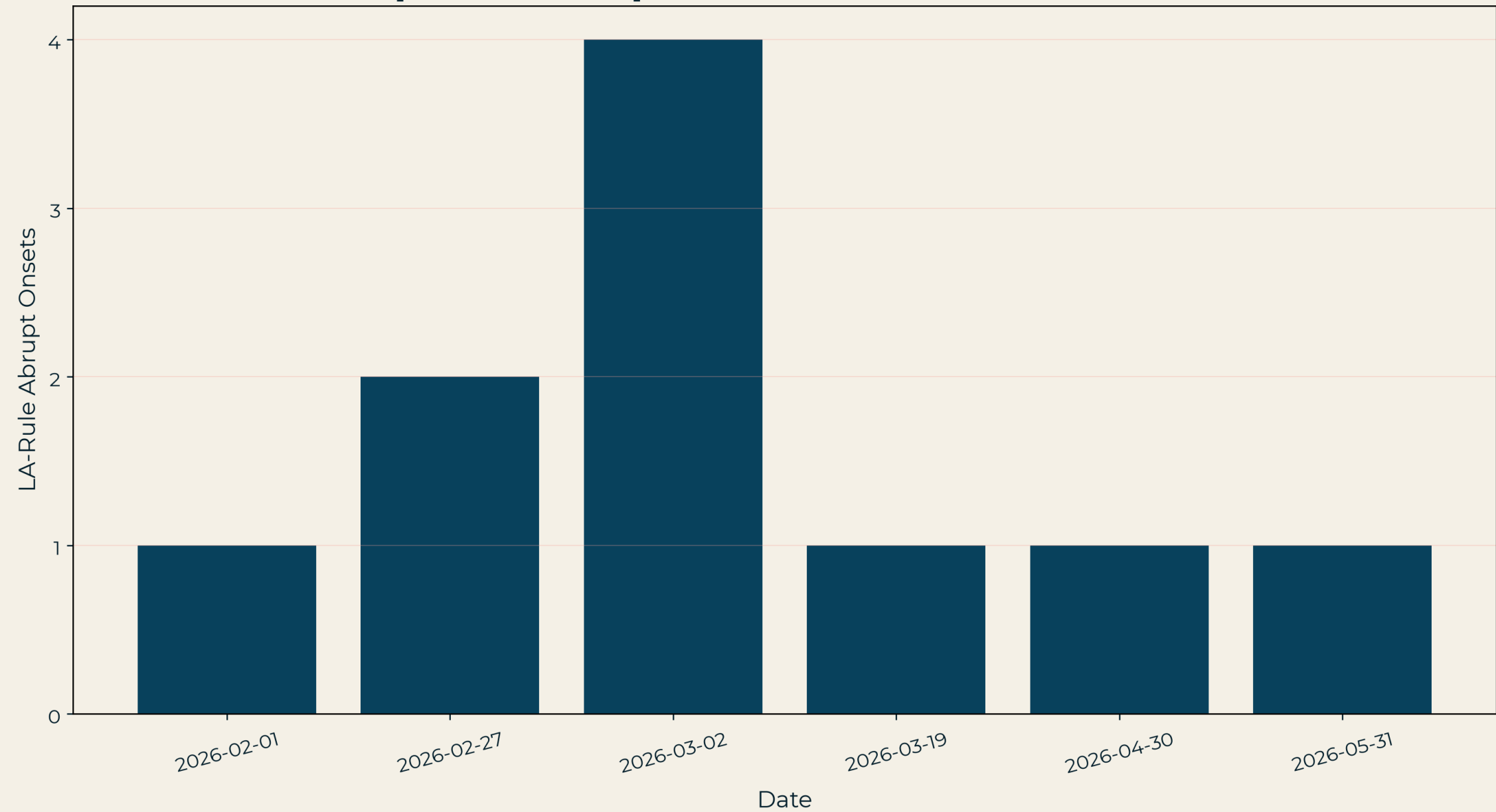
# Midday Congestion Episodes Last Much Longer



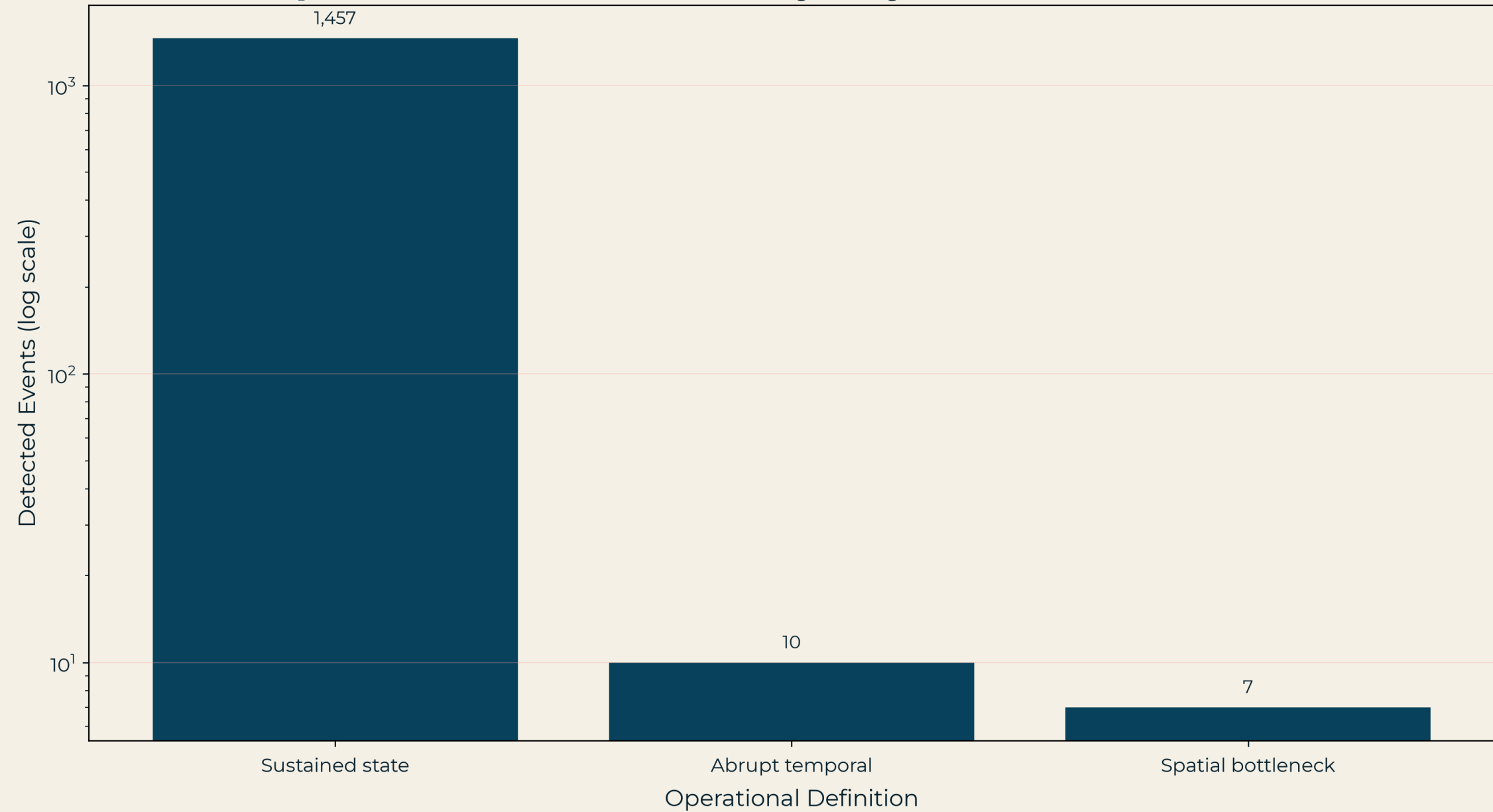
# Headline Duration Results Survive Leave-One-Out Checks



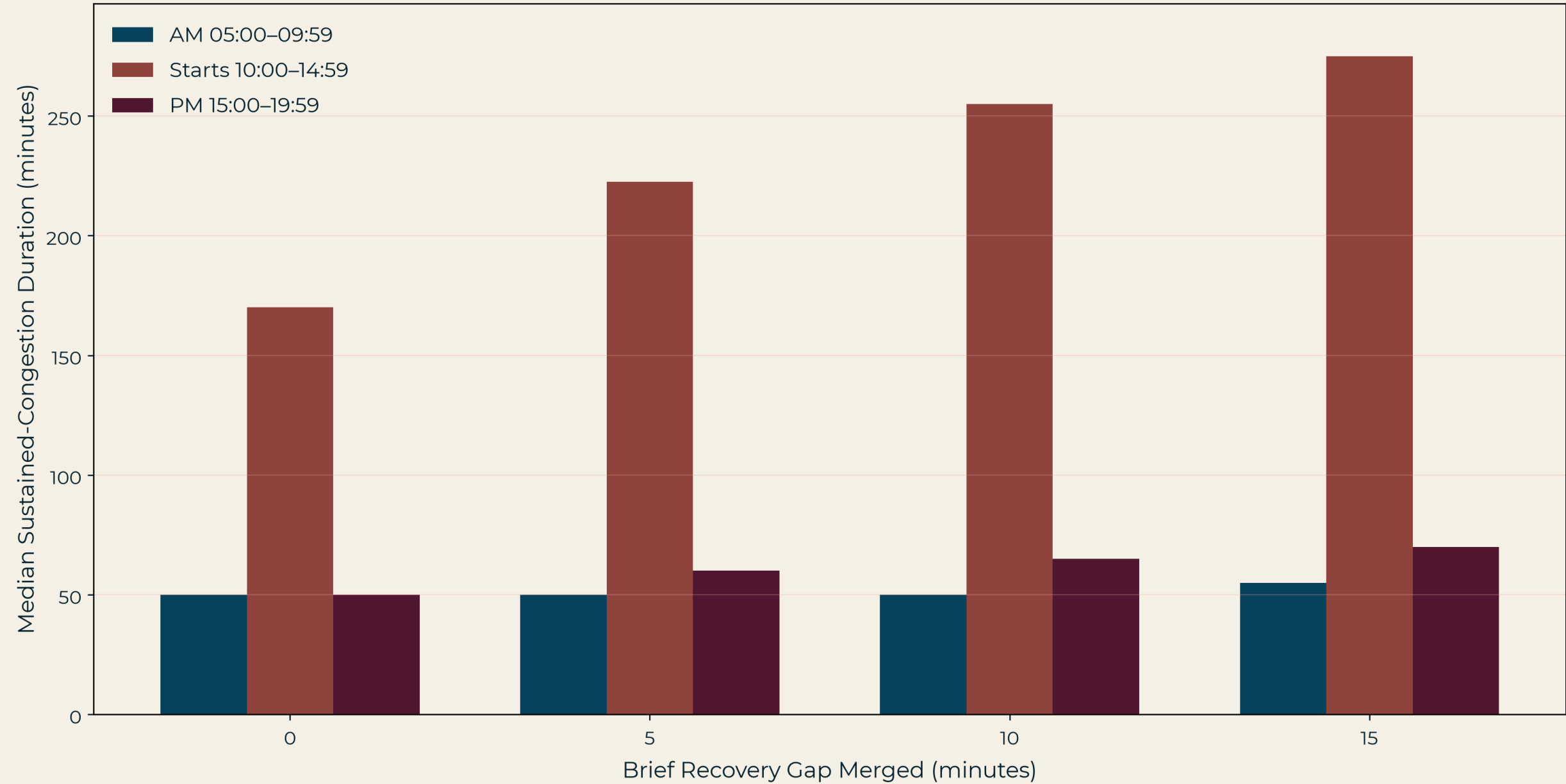
## Abrupt-Onset Sample Is Concentrated on a Few Dates



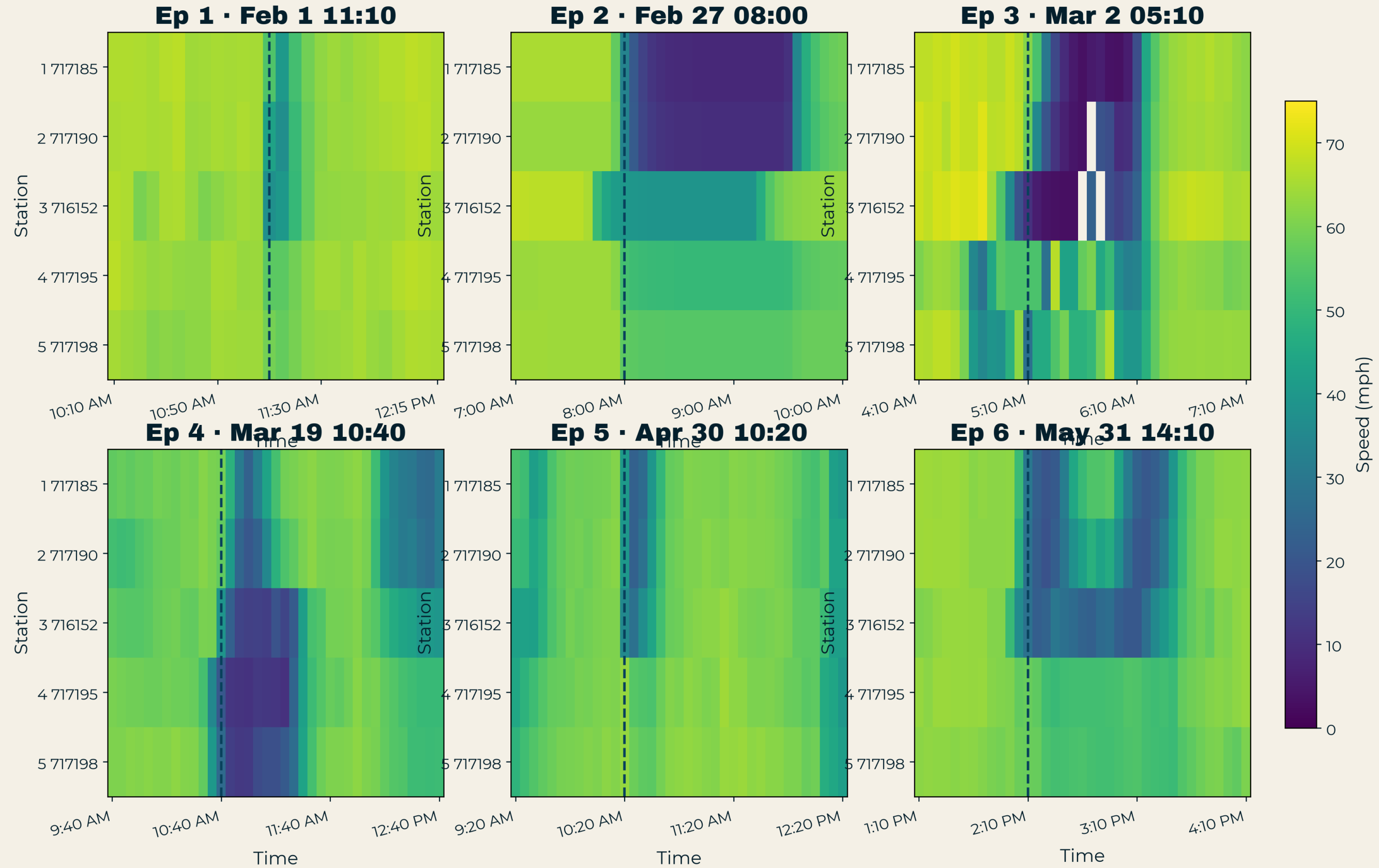
# Operational Definitions Identify Very Different Event Counts



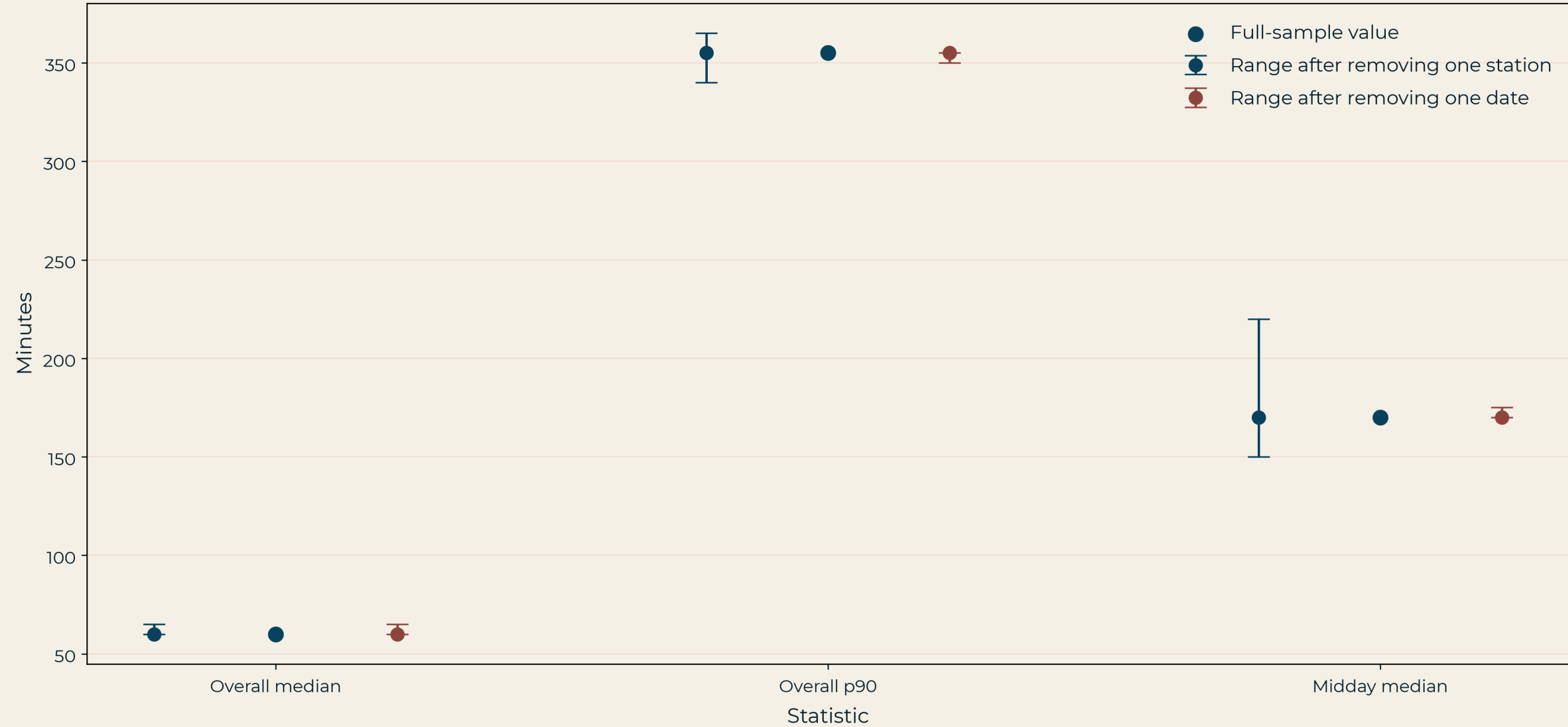
## Episodes Beginning 10:00–15:00 Last Longer



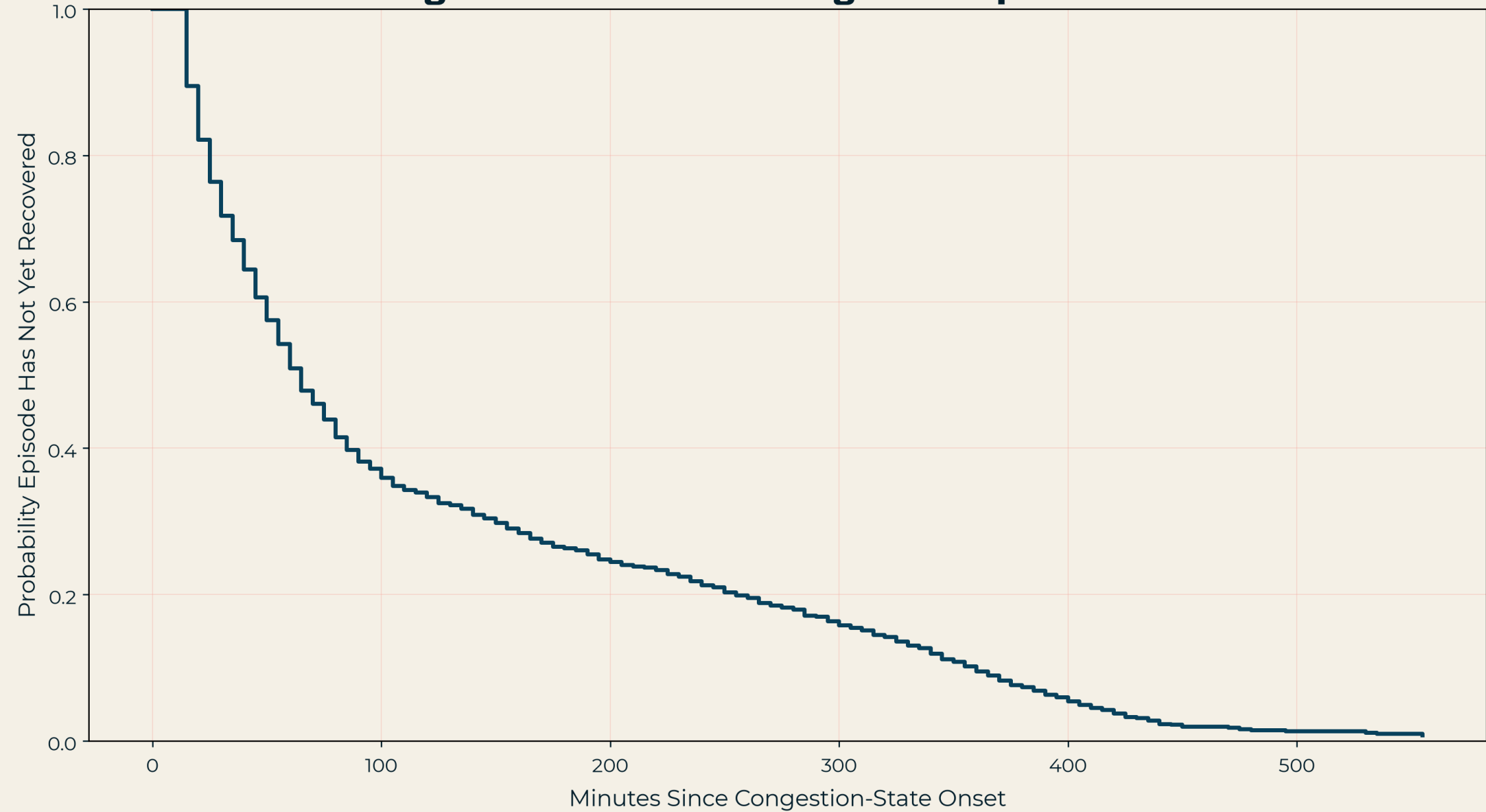
# Speed Evolution Across the Six Corridor Episodes



# Leave-One-Out Sensitivity of Headline Duration Statistics



# How Long Until a Sustained Congestion Episode Recovers?



## Event Counts Remain Similar Across 40–55 mph Thresholds

